

TICVAI

SEAT MANAGEMENT & VENUE MAPPING

Backend Configuration Scope of Work

13 Boards | 130 Configuration Screens | High-Definition UI/UX Reference

Document type Implementation Scope of Work and UI/UX Configuration Reference

Prepared for TICVAI Product & Delivery Team / Dev Implementation Review

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AI-first. Multi-tenant. One authoritative seat inventory.

Document Purpose & Use

Purpose. This document translates the Seat Management & Venue Mapping requirement matrix into thirteen implementation boards. Each board contains one high-definition visual reference and a detailed Scope of Work for its ten backend configuration screens.

Design intent. The visuals communicate information architecture, hierarchy and expected configuration coverage. Development team shall develop production-ready responsive interfaces and may refine interaction details while preserving the approved business scope, shared architecture and TICVAI design system.

Delivery interpretation. A screen is not complete when only the visible mockup is reproduced. The delivered feature must include the described rules, validations, permissions, integrations, states, audit evidence, performance controls and exception handling.

Recommended Shared Seat Architecture

Venue → Map → Layout Version → Performance → Seat/Zone Inventory → Lock/Hold/Reservation/Sale → Ticket/Access → Audit

Architectural requirement. Development team shall implement one shared Seat Inventory Engine used by every TICVAI sales and service channel. Venue editing, customer selection, POS, B2B, call center, partner APIs and reporting must not create independent seat-state logic or separate copies of availability.

Platform-Wide Architectural Requirements

- Authoritative inventory: every seat and standing-zone capacity unit shall have one current state owned by the Seat Inventory Service.
- Atomic concurrency: locks, holds, reservations, sales, releases and exchanges shall use version checks, idempotency and conflict-safe transactions to prevent overselling.
- Multi-tenancy: every applicable screen shall support tenant, brand, region, venue, legal entity, language, timezone, currency and channel context with strict data isolation.
- Shared services: reuse Event, Product, Pricing, Promotion, Cart, Order, Payment, Ticket, Access, Customer, Group, Approval, Notification, Audit and AI services rather than duplicate them.
- Governance: high-impact configuration shall support draft, validation, simulation, review, approval, publication, versioning, rollback and immutable history.
- Accessibility: venue maps, selection, recommendation and administration shall meet WCAG 2.2 AA-aligned interaction requirements and preserve accessible/companion inventory rules.
- AI governance: recognition, design, recommendation, forecasting and pricing outputs shall be explainable, confidence-scored, reviewable and subordinate to business rules and inventory authority.
- Integration: APIs and events shall be secured, versioned and observable and support retries, idempotency, rate limits, correlation IDs, reconciliation and backward-compatible change management.
- Security: access shall use SSO/MFA and RBAC/PBAC with field masking, segregation of duties, temporary/delegated access and controlled emergency override.
- Operational quality: grids require search, filter, sort, pagination, saved views and controlled export; all flows require empty, loading, warning, conflict, error and recovery states.

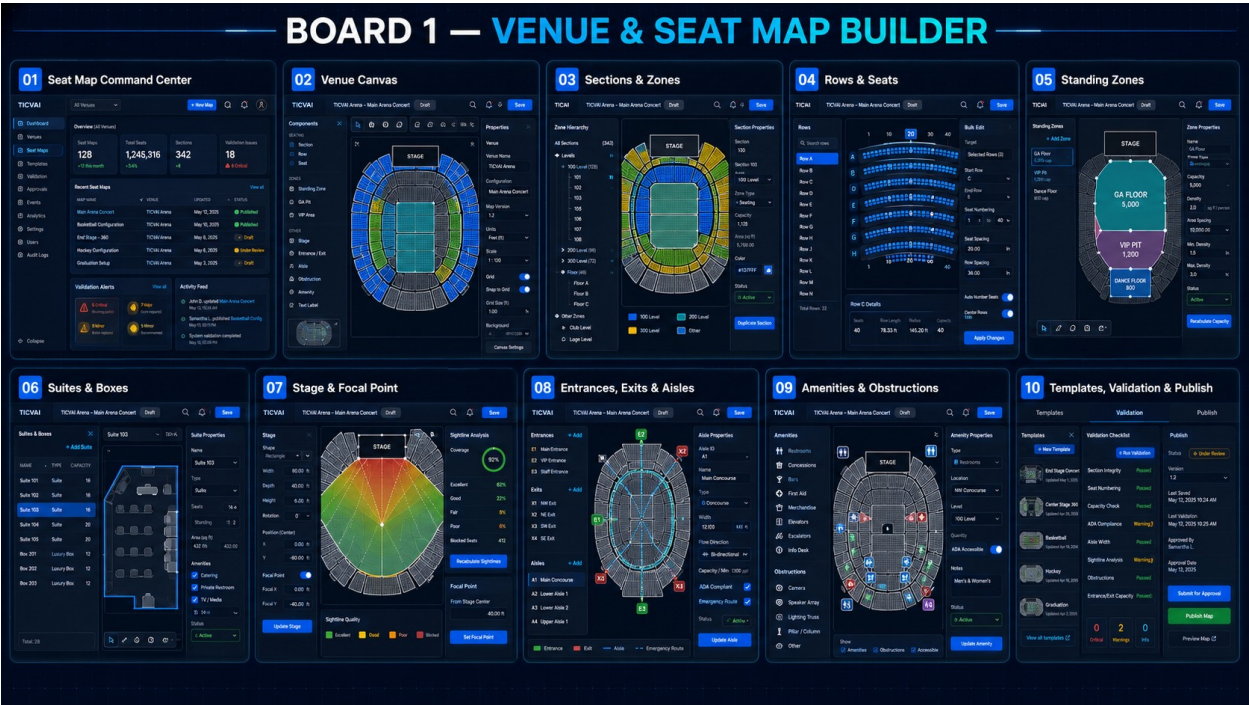
- Performance: map loading, availability refresh, lock acquisition and selection shall be measured against agreed peak-demand SLOs and tested at venue/event scale.
- Auditability: record actor/client, timestamp, tenant/venue/performance, entity, before/after value, rule/model/version, reason, approval, source and correlation ID.

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Board 1 - Venue & Seat Map Builder

Figure 1. High-definition configuration board with all 10 screens.



Visual reference: information architecture and configuration coverage; detailed production behavior is defined in the following scope.

Board 1 Scope of Work

Purpose. Create accurate, reusable venue maps and seating geometry through a governed drag-and-drop administration experience.

Source coverage. Matrix section 21.1 - Seat Map Builder

Operating workflow. Create or select venue -> define scale and focal point -> build sections, rows, seats and zones -> add circulation and facilities -> validate capacity, numbering, accessibility and sightlines -> approve and publish.

Mandatory integrations. Venue Master, Event Management, Product and Pricing, Access Control, Resource Management, Accessibility, Approval, Audit, B2C/B2B/POS channels and Seat Inventory Service.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Seat Map Command Center

Purpose: Provide a role-specific overview of every venue map and its operational readiness.

- Show total venues, active maps, seats, sections, capacity, drafts, approvals, validation errors and upcoming layout changes.
- Filter and compare by tenant, brand, region, venue, map type, status, owner and last-published date.
- Open recent maps, validation alerts, approval tasks and impacted performances directly from the dashboard.
- Version geometry and property changes, enforce venue-scoped permissions, validate affected capacity and preserve before/after audit evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 02 - Venue Canvas

Purpose: Provide the primary drag-and-drop workspace for constructing a venue map.

- Offer components for section, row, seat, standing zone, suite, stage, entrance, exit, aisle, facility, obstruction and label.
- Support zoom, pan, snap-to-grid, rulers, coordinates, layers, alignment, grouping, undo/redo and background-reference controls.
- Configure venue name, dimensions, unit, scale, orientation, origin, focal point and canvas boundaries in a properties inspector.
- Version geometry and property changes, enforce venue-scoped permissions, validate affected capacity and preserve before/after audit evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 03 - Sections & Zones

Purpose: Define the commercial and operational hierarchy of the venue.

- Create, draw and edit section or zone polygons with name, code, level, capacity, color, category and parent hierarchy.
- Support curved, rectangular, freeform and imported boundaries with duplication, alignment and bulk-property updates.

- Map sections to pricing, access gates, sales channels, accessibility attributes and reporting dimensions.
- Version geometry and property changes, enforce venue-scoped permissions, validate affected capacity and preserve before/after audit evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 04 - Rows & Seats

Purpose: Build and maintain numbered rows and individual seat positions at scale.

- Generate straight, curved, radial or custom rows using seat count, spacing, radius, angle, direction and offset parameters.
- Configure seat number, label, type, category, coordinate, status default, accessibility, view quality and amenity attributes.
- Support bulk add, renumber, reverse, insert, remove, copy, align and spacing changes with collision and duplicate detection.
- Version geometry and property changes, enforce venue-scoped permissions, validate affected capacity and preserve before/after audit evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 05 - Standing Zones

Purpose: Configure non-assigned areas with controlled capacity and density.

- Draw standing, general-admission, pit, dance-floor or hospitality zones and assign type, capacity, density and color.
- Calculate capacity from area and approved density while allowing an authorized lower operating limit.
- Associate entrances, exits, age rules, access products, pricing bands and event-specific restrictions.
- Version geometry and property changes, enforce venue-scoped permissions, validate affected capacity and preserve before/after audit evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 06 - Suites & Boxes

Purpose: Model suites, boxes and hospitality spaces as sellable seating inventory.

- Create suites and boxes with code, capacity, internal seat layout, standing allowance, amenities and accessibility attributes.
- Support whole-suite, per-seat, shared and configurable sales models with linked products and price categories.
- Maintain owner, contract, allocation, entrance, service area and operational-status references without duplicating CRM or contract data.
- Version geometry and property changes, enforce venue-scoped permissions, validate affected capacity and preserve before/after audit evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 07 - Stage & Focal Point

Purpose: Define the viewing focal point and production footprint used by seating logic.

- Add stage, field, screen, court, track or custom focal areas with position, dimensions, rotation, height and event-specific variant.
- Configure one or multiple focal points for distance, orientation, best-seat and closest-to-stage calculations.
- Preview sightline coverage, obstructed areas and seats affected by production structures before publication.
- Version geometry and property changes, enforce venue-scoped permissions, validate affected capacity and preserve before/after audit evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 08 - Entrances, Exits & Aisles

Purpose: Map circulation elements used by access, accessibility and safety operations.

- Create entrances, exits, gates, portals, vomitories, concourses, stairways and aisles with unique identifiers and attributes.
- Define width, direction, level, connected zones, access-control device, accessible status and emergency designation.
- Validate disconnected zones, blocked paths, minimum aisle width and route continuity against configured venue rules.
- Version geometry and property changes, enforce venue-scoped permissions, validate affected capacity and preserve before/after audit evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 09 - Amenities & Obstructions

Purpose: Add guest facilities, service points and view-affecting objects to the venue map.

- Place restrooms, concessions, bars, first aid, merchandise, lifts, escalators, information, prayer and accessibility facilities.
- Map pillars, cameras, speaker arrays, lighting trusses, barriers and production structures with obstruction type and dimensions.
- Expose approved amenity and view attributes to filters, seat previews, operations and accessibility routing.
- Version geometry and property changes, enforce venue-scoped permissions, validate affected capacity and preserve before/after audit evidence.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 10 - Templates, Validation & Publish

Purpose: Govern reusable venue templates and release a map only when it is operationally complete.

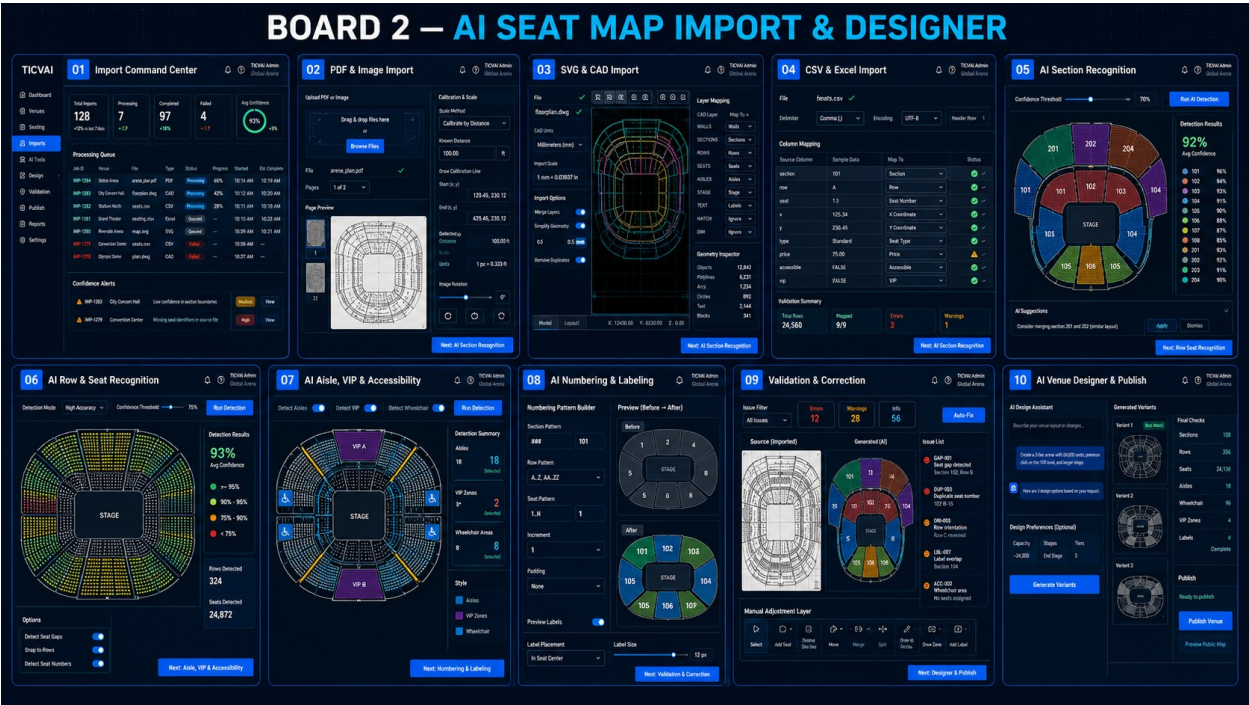
- Create, classify, clone, archive and reuse venue templates while preserving lineage and ownership.
- Run checks for geometry, capacity, duplicate numbering, overlaps, missing routes, accessibility, sightlines and dependent products.
- Provide draft, review, approval, scheduled publication, preview, version comparison and rollback with impact confirmation.

- Version geometry and property changes, enforce venue-scoped permissions, validate affected capacity and preserve before/after audit evidence.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.*

Board 2 - AI Seat Map Import & Designer

Figure 2. High-definition configuration board with all 10 screens.



Visual reference: information architecture and configuration coverage; detailed production behavior is defined in the following scope.

Board 2 Scope of Work

Purpose. Convert PDF, image, vector, CAD and tabular venue files into validated TICVAI seat maps through explainable AI and controlled manual correction.

Source coverage. Matrix section 21.2 - Seat Map Import

Operating workflow. Upload source -> classify and calibrate -> map layers or fields -> recognize sections, rows, seats, aisles and special areas -> number and label -> review confidence and issues -> correct -> validate -> approve and publish.

Mandatory integrations. Document Storage, Malware Scanning, AI Model Service, Venue Canvas, Seat Inventory, Accessibility, Approval, Audit, Notification and secured import APIs.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Import Command Center

Purpose: Monitor every seat-map import from submission to publication.

- Show queued, processing, completed, failed and review-required jobs with source format, venue, owner and elapsed time.
- Display average confidence, issue counts, model version and jobs blocked by malware, unsupported content or validation failures.
- Support retry, cancel, duplicate, assign reviewer, open result and download controlled error reports.
- Retain source files, model/version, confidence, reviewer corrections and publication evidence; never allow AI output to bypass validation or approval.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 02 - PDF & Image Import

Purpose: Ingest PDF, PNG, JPG and supported scanned map sources.

- Provide drag-and-drop upload, page selection, preview, crop, rotation, de-skew, contrast and noise-reduction controls.
- Calibrate scale using known distance, dimensions or reference objects and capture venue orientation and focal point.
- Require file validation, malware scanning, size limits and a clear unsupported-or-low-quality recovery path.
- Retain source files, model/version, confidence, reviewer corrections and publication evidence; never allow AI output to bypass validation or approval.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 03 - SVG & CAD Import

Purpose: Convert vector and CAD geometry while preserving useful source layers.

- Support approved SVG, DXF and DWG workflows with unit, scale, origin, coordinate and rotation mapping.
- Map source layers to sections, rows, seats, aisles, stage, text, amenities, obstructions and ignored content.
- Simplify excessive geometry, close open paths, remove duplicates and preview the transformed result before recognition.

- Retain source files, model/version, confidence, reviewer corrections and publication evidence; never allow AI output to bypass validation or approval.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 04 - CSV & Excel Import

Purpose: Build maps or update seat master data from structured rows.

- Map columns for venue, level, section, row, seat, coordinates, type, category, price band, accessibility and status.
- Configure delimiter, header, encoding, sheet, data type, defaults and transformation rules with a sample-data preview.
- Detect missing identifiers, duplicates, invalid coordinates, inconsistent row sequences and unmapped values before import.
- Retain source files, model/version, confidence, reviewer corrections and publication evidence; never allow AI output to bypass validation or approval.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 05 - AI Section Recognition

Purpose: Recognize venue levels, sections and standing zones from source geometry.

- Display detected boundaries, labels, hierarchy, capacity estimate and per-object confidence on the source preview.
- Allow accept, reject, merge, split, rename and redraw with AI correction suggestions and reason capture.
- Flag ambiguous, overlapping, disconnected or unlabelled areas and prevent silent assignment to the wrong level.
- Retain source files, model/version, confidence, reviewer corrections and publication evidence; never allow AI output to bypass validation or approval.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 06 - AI Row & Seat Recognition

Purpose: Detect rows and individual seat positions at production scale.

- Recognize row curves, direction, gaps, seat dots, repeated symbols and wheelchair spaces with confidence heat maps.
- Allow threshold, gap, snap, spacing and symbol-class controls followed by selective reprocessing.
- Present detected counts by section and reconcile them with stated capacities, labels and expected patterns.
- Retain source files, model/version, confidence, reviewer corrections and publication evidence; never allow AI output to bypass validation or approval.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 07 - AI Aisle, VIP & Accessibility

Purpose: Recognize special geometry and operational attributes that affect seating.

- Detect aisles, entrances, exits, stages, VIP areas, suites, boxes, wheelchair positions and accessible routes.
- Show confidence, source evidence and conflicts where a detected object overlaps inventory or breaks route continuity.
- Allow the reviewer to change object class, boundary and attributes without restarting the full import.

- Retain source files, model/version, confidence, reviewer corrections and publication evidence; never allow AI output to bypass validation or approval.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 08 - AI Numbering & Labeling

Purpose: Generate consistent section, row and seat identifiers from recognized geometry.

- Configure alphabetic, numeric, alphanumeric, odd/even, continuous, reset-by-row and venue-specific numbering patterns.
- Preview before/after labels, direction, padding, skipped values, reserved labels and accessible-seat suffixes.
- Detect duplicates, gaps, reversals and conflicts with existing venue identifiers before applying changes.
- Retain source files, model/version, confidence, reviewer corrections and publication evidence; never allow AI output to bypass validation or approval.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 09 - Validation & Correction

Purpose: Provide one controlled workspace for resolving AI and import exceptions.

- Compare the source and generated map side by side with issue list, filters, severity and confidence.
- Offer manual draw, move, split, merge, add, delete, renumber, relabel and property-edit tools on a correction layer.
- Re-run selected recognizers, validate affected objects and preserve every AI suggestion, human edit and reviewer decision.
- Retain source files, model/version, confidence, reviewer corrections and publication evidence; never allow AI output to bypass validation or approval.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 10 - AI Venue Designer & Publish

Purpose: Generate or refine a seat map from a natural-language brief and release the approved result.

- Accept venue type, dimensions, levels, target capacity, stage, section mix, accessibility and circulation requirements.
- Generate multiple explainable variants with capacity, sightline, accessible inventory and constraint summaries.
- Require full map validation, human review, approval and controlled one-click publication with version and rollback.
- Retain source files, model/version, confidence, reviewer corrections and publication evidence; never allow AI output to bypass validation or approval.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Board 3 - Layouts, Templates & Versions

Figure 3. High-definition configuration board with all 10 screens.



Visual reference: information architecture and configuration coverage; detailed production behavior is defined in the following scope.

Board 3 Scope of Work

Purpose. Create reusable and event-specific seating layouts while protecting sold inventory through versioning, impact analysis and controlled publication.

Source coverage. Matrix section 21.3 - Layout & Templates

Operating workflow. Select base map or template -> clone or create event layout -> configure production changes -> assign to performances -> simulate impact -> block or release seats -> review and approve -> publish or roll back.

Mandatory integrations. Venue Maps, Event and Performance Management, Product and Pricing, Seat Inventory, Orders, Holds, Access Control, Approval, Notification, Audit and Reporting.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Layout Command Center

Purpose: Manage layouts, versions and performance assignments from one operational view.

- Show active layouts, draft versions, total capacity, utilization, upcoming changes, validation issues and approval status.
- List layouts by venue, template, event, performance, version, owner, status and effective period.
- Surface urgent sold-seat impact, missing assignment and publication conflicts with direct remediation links.
- Preserve version lineage and require impact analysis for every change affecting capacity, products, sold seats, holds, access routes or pricing.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 02 - Template Library

Purpose: Maintain reusable layouts for common venue configurations.

- Provide templates for end-stage, center-stage, sports, theatre, banquet, classroom and venue-defined patterns.
- Store capacity, focal point, production footprint, section availability, access routes and default blocked inventory.
- Support preview, clone, compare, archive, ownership and controlled sharing across authorized tenants or venues.
- Preserve version lineage and require impact analysis for every change affecting capacity, products, sold seats, holds, access routes or pricing.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 03 - Event-Specific Layout

Purpose: Create a seating layout tailored to one event or production.

- Start from the current venue map or approved template and adjust stage, field, sections, rows, seats, aisles and facilities.
- Display live capacity by section/category and affected pricing, products, access gates and operational resources.
- Support drafts, notes, attachments, production requirements and validation before performance assignment.
- Preserve version lineage and require impact analysis for every change affecting capacity, products, sold seats, holds, access routes or pricing.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 04 - Clone & Inheritance

Purpose: Reuse a layout without losing the relationship to its approved source.

- Clone a layout for another event, venue or performance and select which settings are inherited, copied or excluded.
- Show inherited versus overridden geometry, categories, holds, accessibility, pricing and production settings.
- Warn when later source changes could affect dependent clones and require explicit synchronization decisions.
- Preserve version lineage and require impact analysis for every change affecting capacity, products, sold seats, holds, access routes or pricing.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 05 - Version Compare

Purpose: Review exact differences between layout versions before approval or rollback.

- Provide side-by-side and overlay comparisons of added, removed, moved and changed sections, rows, seats and routes.
- Summarize capacity, accessible inventory, pricing, hold, product and revenue impact by category.
- Link each change to actor, timestamp, request, reason, approval and impacted performance.
- Preserve version lineage and require impact analysis for every change affecting capacity, products, sold seats, holds, access routes or pricing.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 06 - Multi-Performance Assignment

Purpose: Assign approved layouts to one or many event performances.

- Provide calendar and grid views of event, date, venue, performance, assigned layout/version and status.
- Support bulk assignment, date range, recurrence, exception dates and different layouts across performances.
- Prevent assignment when a performance has incompatible sales, capacity, access or already-issued inventory.
- Preserve version lineage and require impact analysis for every change affecting capacity, products, sold seats, holds, access routes or pricing.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 07 - Temporary Seat Blocking

Purpose: Temporarily remove seats from sale for production or operational reasons.

- Select section, row, seat or polygon on the map and define block type, reason, owner and effective period.
- Preview affected available, held, reserved and sold inventory with required remediation or approval.
- Support camera, equipment, safety, maintenance, sightline, house and event-production block categories.
- Preserve version lineage and require impact analysis for every change affecting capacity, products, sold seats, holds, access routes or pricing.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 08 - Scheduled Seat Release

Purpose: Return temporarily blocked inventory to sale at a controlled time.

- Create release schedules by event, performance, hold/block type, seat set, time before event and condition.
- Preview release quantity, new capacity, pricing/category mapping and affected sales channels.
- Support reschedule, cancel, manual run, partial failure recovery and stakeholder notifications.
- Preserve version lineage and require impact analysis for every change affecting capacity, products, sold seats, holds, access routes or pricing.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 09 - Conflict & Impact Simulation

Purpose: Identify operational and commercial risk before a layout change becomes effective.

- Simulate seat removal, addition, movement, category change, stage change, route change and block/release actions.
- Calculate affected sold seats, holds, reservations, carts, products, access gates, capacity and forecast revenue.
- Provide resolution options such as reseating, alternative inventory, delayed release, refund workflow or rejected change.
- Preserve version lineage and require impact analysis for every change affecting capacity, products, sold seats, holds, access routes or pricing.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 10 - Approval, Publish & Rollback

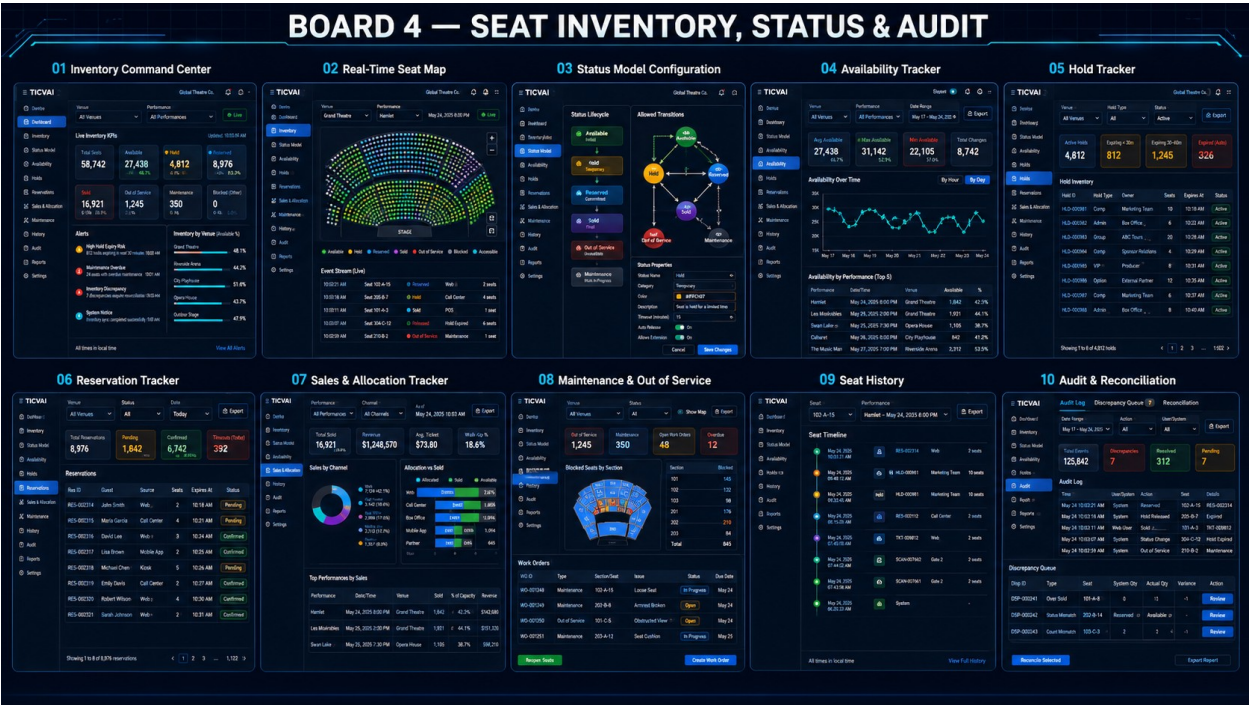
Purpose: Govern the complete release lifecycle for layout changes.

- Configure draft, review, operations, ticketing, finance and final approval stages by impact and venue policy.
- Schedule publication, notify dependent systems and verify synchronized version across channels.
- Maintain version history, failed-publication recovery, emergency unpublish and safe rollback with impact confirmation.
- Preserve version lineage and require impact analysis for every change affecting capacity, products, sold seats, holds, access routes or pricing.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Board 4 - Seat Inventory, Status & Audit

Figure 4. High-definition configuration board with all 10 screens.



Visual reference: information architecture and configuration coverage; detailed production behavior is defined in the following scope.

Board 4 Scope of Work

Purpose. Provide the authoritative, real-time operational record of every seat status, availability change, hold, reservation, sale and maintenance event.

Source coverage. Matrix section 21.4 - Seat Tracking & Holds

Operating workflow. Receive inventory event -> validate version and transition -> update authoritative seat state atomically -> broadcast availability -> reconcile channels -> investigate history or discrepancy -> correct through controlled action.

Mandatory integrations. Seat Map and Layout, Event/Performance, Product and Pricing, Cart, Orders, Reservations, Holds, POS/B2C/B2B, Access Control, Maintenance, Event Bus, Audit and Finance Reconciliation.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Inventory Command Center

Purpose: Monitor live seat inventory health across all venues and performances.

- Show total, available, held, reserved, sold, blocked, out-of-service, accessible and discrepancy counts.
- Filter by tenant, venue, event, performance, layout, section, category, channel and time window.
- Surface expiring locks, hold spikes, maintenance impact, synchronization lag and oversell risk.
- Use atomic transitions, optimistic concurrency, idempotent events and immutable history; historical seat facts cannot be silently overwritten.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 02 - Real-Time Seat Map

Purpose: Display the current authoritative status of each seat and zone.

- Render available, locked, held, reserved, sold, blocked, out-of-service and accessible states with an interactive legend.
- Stream near-real-time state changes with source, transaction reference and event timestamp.
- Allow authorized drill-down to seat details without permitting unsafe manual state changes.
- Use atomic transitions, optimistic concurrency, idempotent events and immutable history; historical seat facts cannot be silently overwritten.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 03 - Status Model Configuration

Purpose: Define the permitted seat lifecycle and transition rules.

- Maintain status code, label, color, terminal flag, availability effect, channel visibility and inventory treatment.
- Configure allowed transitions, required reason, approval, timeout, event source and compensating action.
- Simulate changes and prohibit circular, orphaned or ambiguous transitions before activation.
- Use atomic transitions, optimistic concurrency, idempotent events and immutable history; historical seat facts cannot be silently overwritten.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 04 - Availability Tracker

Purpose: Analyze current and historical seat availability.

- Show available quantity by venue, performance, section, row, category, price band and sales channel.
- Graph availability over time and identify material changes caused by locks, holds, reservations, sales or blocks.
- Provide secured export and links to the exact transactions or events underlying each change.
- Use atomic transitions, optimistic concurrency, idempotent events and immutable history; historical seat facts cannot be silently overwritten.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 05 - Hold Tracker

Purpose: Track active and historical held inventory regardless of hold type.

- List hold ID, type, owner, event, section/seat set, quantity, created time, expiry, priority and status.
- Filter upcoming expiry, extended, converted, released and conflicting holds and open the governing rule.
- Allow approved release, extend or reassign actions while preserving the original hold record.
- Use atomic transitions, optimistic concurrency, idempotent events and immutable history; historical seat facts cannot be silently overwritten.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 06 - Reservation Tracker

Purpose: Monitor seat reservations and temporary allocations before final sale.

- List reservation, guest/group, source, seat set, lock/hold reference, expiry, payment state and confirmation status.
- Identify abandoned, timed-out, payment-pending, orphaned and channel-mismatched reservations.
- Support controlled retry, release, re-link or escalation without bypassing payment or inventory authority.
- Use atomic transitions, optimistic concurrency, idempotent events and immutable history; historical seat facts cannot be silently overwritten.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 07 - Sales & Allocation Tracker

Purpose: Trace sold seats and inventory allocation across channels and partners.

- Report seats and revenue by web, mobile, POS, call center, box office, B2B reseller and partner API.
- Show allocation-versus-sold performance, remaining quota, order, ticket, price, fee and fulfillment status.
- Detect duplicate issue, invalid allocation, reversed payment and seat/order inconsistency.
- Use atomic transitions, optimistic concurrency, idempotent events and immutable history; historical seat facts cannot be silently overwritten.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 08 - Maintenance & Out of Service

Purpose: Control seats unavailable because of defects, inspection or venue work.

- Create or link work orders with seat/area, issue, severity, status, owner, due date and supporting evidence.
- Block affected inventory, identify sold/reserved seats and trigger reseating or guest-service workflows.
- Return seats to available only after required repair, inspection, approval and layout validation.
- Use atomic transitions, optimistic concurrency, idempotent events and immutable history; historical seat facts cannot be silently overwritten.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 09 - Seat History

Purpose: Present the complete chronological history of one seat for one performance or physical venue position.

- Show locks, holds, reservations, sales, releases, refunds, exchanges, scans, blocks and maintenance events.
- Preserve business timestamp, processing timestamp, source, actor, correlation ID, rule/version and related object.
- Provide authorized links to order, ticket, guest, hold, approval and device records with field masking.
- Use atomic transitions, optimistic concurrency, idempotent events and immutable history; historical seat facts cannot be silently overwritten.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 10 - Audit & Reconciliation

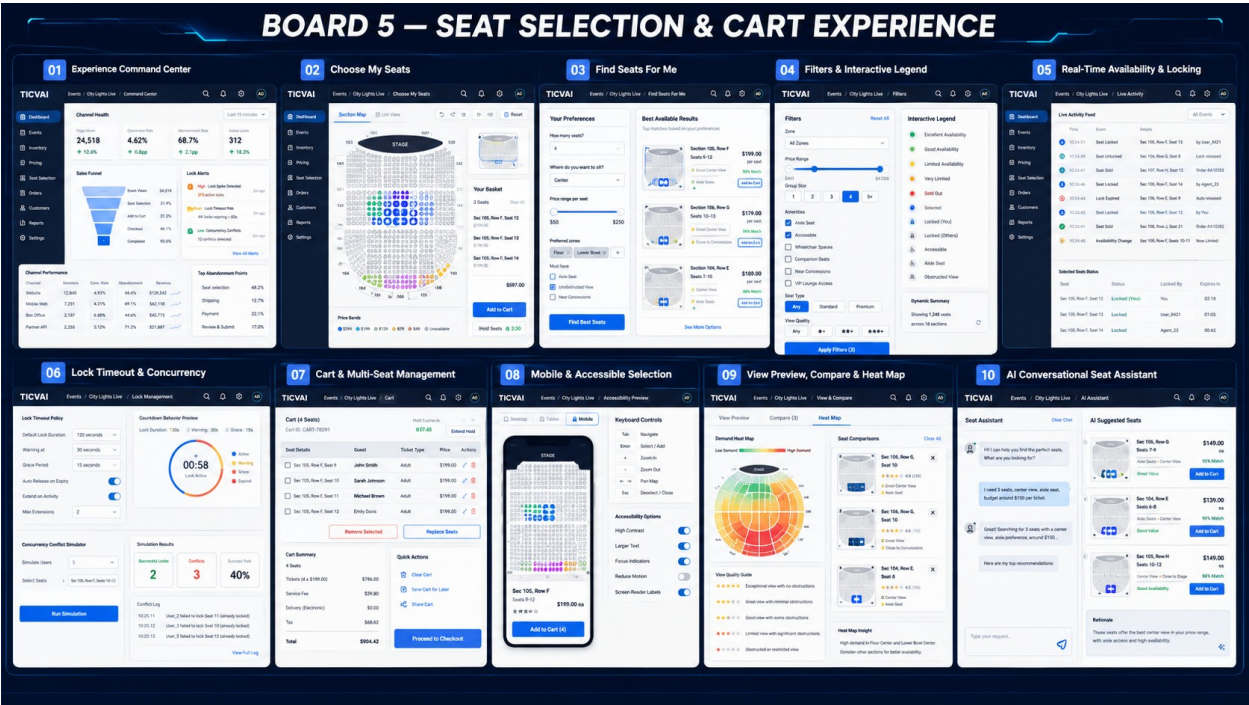
Purpose: Detect and resolve differences between authoritative inventory and connected systems.

- Compare state and quantity across Seat Inventory, Cart, Order, Payment, Ticket, Channel and Access systems.
- Queue discrepancies by severity, age, venue, performance, type and suspected source with recommended action.
- Require reason and approval for corrections, use compensating events and retain complete before/after evidence.
- Use atomic transitions, optimistic concurrency, idempotent events and immutable history; historical seat facts cannot be silently overwritten.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Board 5 - Seat Selection & Cart Experience

Figure 5. High-definition configuration board with all 10 screens.



Visual reference: information architecture and configuration coverage; detailed production behavior is defined in the following scope.

Board 5 Scope of Work

Purpose. Provide fast, accessible and concurrency-safe assigned-seat selection across web, mobile, POS, call center and partner journeys.

Source coverage. Matrix section 21.5 - Seat Selection & Assignment

Operating workflow. Load live performance inventory -> apply filters or preferences -> select or recommend seats -> acquire atomic lock -> add multiple seats to cart -> manage timeout and conflicts -> confirm through checkout -> release unused inventory.

Mandatory integrations. Seat Inventory, Event/Performance, Pricing, Promotions, Cart, Checkout, Payments, Customer/Group Profiles, Accessibility, B2C/B2B/POS/Call Center, Analytics and AI Recommendation Service.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Experience Command Center

Purpose: Monitor seat-selection performance, availability quality and conversion across channels.

- Show map loads, searches, selections, locks, expiries, carts, checkout conversion, abandonment and revenue.
- Compare web, mobile, POS, call center, box office, B2B and API channels by venue and performance.
- Surface slow maps, stale availability, lock conflicts, high timeout and selection drop-off with drill-down.
- All channels shall use the same authoritative availability, lock and pricing services; no channel may maintain an independent seat-state cache beyond approved freshness rules.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.*

Screen 02 - Choose My Seats

Purpose: Allow a guest or authorized agent to select exact seats from an interactive map.

- Display section, row, seat, price, fees, type, view quality, accessibility, amenity and live availability.
- Support zoom, pan, section drill-down, keyboard navigation, clear selection and a live basket summary.
- Acquire locks only after server confirmation and clearly distinguish selected, unavailable, locked and accessible inventory.
- All channels shall use the same authoritative availability, lock and pricing services; no channel may maintain an independent seat-state cache beyond approved freshness rules.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.*

Screen 03 - Find Seats For Me

Purpose: Return best-available seat groups from stated customer preferences.

- Capture quantity, zone, level, price range, accessibility, aisle, view, amenity and proximity preferences.
- Rank valid contiguous or near-contiguous groups with total price, view summary, match score and trade-offs.
- Allow selection, alternative search and return to exact map without losing the current cart context.
- All channels shall use the same authoritative availability, lock and pricing services; no channel may maintain an independent seat-state cache beyond approved freshness rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 04 - Filters & Interactive Legend

Purpose: Make large and complex seat maps understandable and searchable.

- Provide zone, price, group size, amenities, seat type, accessibility, view quality, level and availability filters.
- Update results, counts, map emphasis and legend in real time without hiding selected seats unexpectedly.
- Configure tenant/venue labels and colors while maintaining accessible contrast and non-color status cues.
- All channels shall use the same authoritative availability, lock and pricing services; no channel may maintain an independent seat-state cache beyond approved freshness rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 05 - Real-Time Availability & Locking

Purpose: Show and control atomic seat locks during selection.

- Display lock owner class, channel, transaction, acquisition time, expiry and current authoritative state to permitted roles.
- Broadcast lock, unlock, sale, hold, block and availability events to active selection sessions.
- Handle lock rejection, partial group success, lost connectivity and stale version with clear recovery options.
- All channels shall use the same authoritative availability, lock and pricing services; no channel may maintain an independent seat-state cache beyond approved freshness rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 06 - Lock Timeout & Concurrency

Purpose: Configure fair timeout and collision behavior under concurrent demand.

- Set default lock duration, warning, grace period, extension policy, maximum extension and auto-release by channel or event.
- Configure optimistic concurrency, version checks, idempotency, conflict messages and retry boundaries.
- Simulate multiple customers selecting the same inventory and verify that no double lock or oversell can occur.
- All channels shall use the same authoritative availability, lock and pricing services; no channel may maintain an independent seat-state cache beyond approved freshness rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 07 - Cart & Multi-Seat Management

Purpose: Manage several seats as one controlled cart allocation.

- List each seat, assigned guest, ticket type, price, fee, promotion, lock expiry and eligibility status.
- Support add, remove, replace, reassign, save-for-later where permitted and clear-cart actions.
- Revalidate availability, price, eligibility and adjacency before checkout and release all abandoned locks reliably.
- All channels shall use the same authoritative availability, lock and pricing services; no channel may maintain an independent seat-state cache beyond approved freshness rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 08 - Mobile & Accessible Selection

Purpose: Deliver an equivalent and WCAG-aligned selection flow on supported devices.

- Provide responsive map/list modes, large targets, screen-reader labels, logical focus order, keyboard and switch access.
- Support high contrast, text scaling, reduced motion and clear wheelchair, companion, aisle and route information.
- Maintain selection and lock state through rotation, app backgrounding, network interruption and session recovery.
- All channels shall use the same authoritative availability, lock and pricing services; no channel may maintain an independent seat-state cache beyond approved freshness rules.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.*

Screen 09 - View Preview, Compare & Heat Map

Purpose: Help customers understand qualitative and demand differences before choosing.

- Show approved seat-view previews or representative section views with source, freshness and obstruction disclosure.
- Compare seats by price, distance, view, level, amenities, accessibility and overall match.
- Display configurable demand, price or availability heat maps without revealing sensitive sales strategy.
- All channels shall use the same authoritative availability, lock and pricing services; no channel may maintain an independent seat-state cache beyond approved freshness rules.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.*

Screen 10 - AI Conversational Seat Assistant

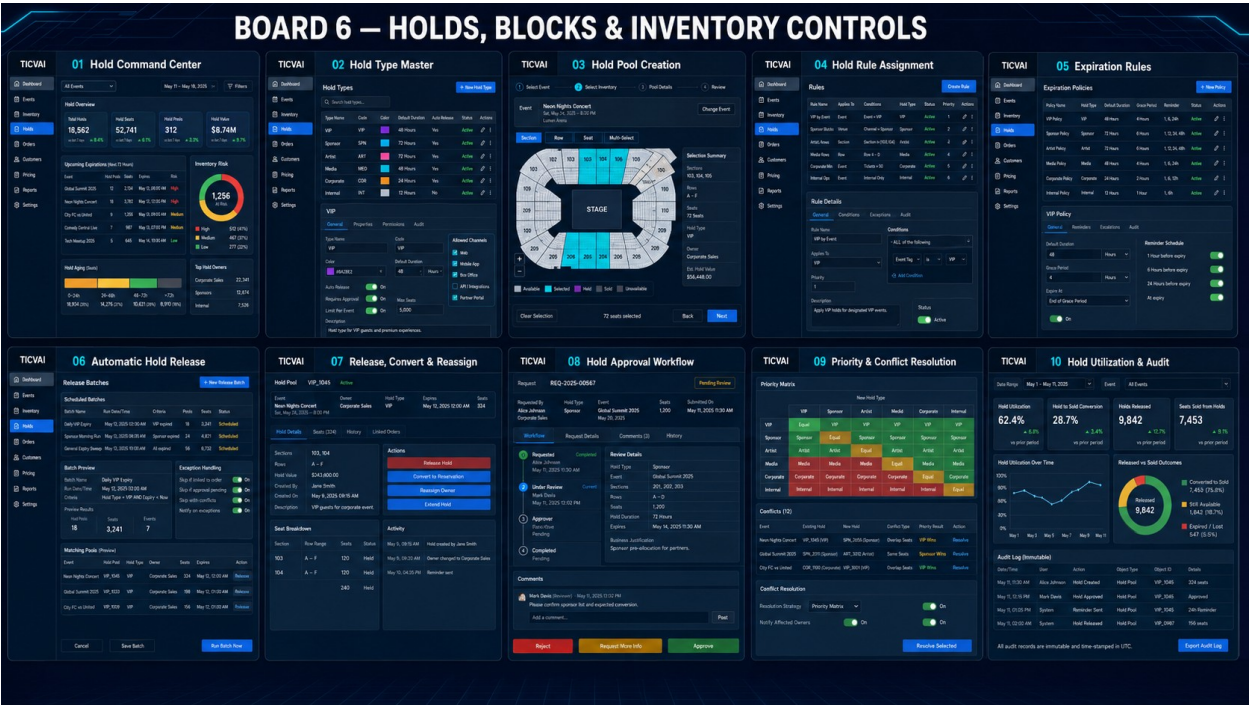
Purpose: Allow natural-language seat discovery through governed AI.

- Understand requests such as family size, budget, view, aisle, stage proximity, accessibility and preferred level.
- Return only currently valid seat groups with match score, rationale, trade-offs, total price and add-to-cart action.
- Record model/version and feedback, disclose uncertainty and prevent AI from bypassing locks, pricing, eligibility or accessible-seat rules.
- All channels shall use the same authoritative availability, lock and pricing services; no channel may maintain an independent seat-state cache beyond approved freshness rules.

Acceptance condition: *Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.*

Board 6 - Holds, Blocks & Inventory Controls

Figure 6. High-definition configuration board with all 10 screens.



Visual reference: information architecture and configuration coverage; detailed production behavior is defined in the following scope.

Board 6 Scope of Work

Purpose. Create and govern VIP, sponsor, artist, media, corporate and internal holds with clear ownership, expiry, release and commercial accountability.

Source coverage. Matrix section 21.6 - Seat Holds & Blocks

Operating workflow. Define hold type -> select inventory -> apply event/stakeholder rule -> detect priority conflict -> request and approve -> activate -> notify before expiry -> release, convert, extend or reassign -> measure utilization and audit.

Mandatory integrations. Seat Inventory, Event/Performance, CRM and B2B Accounts, Reservations, Orders, Contracts, Pricing, Approval, Notification, Reporting and Audit.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Hold Command Center

Purpose: Provide a complete operational view of held inventory and its risk.

- Show held seats, hold value, utilization, conversion, upcoming expiry, overdue decisions and inventory at risk.
- Analyze by tenant, venue, event, performance, hold type, owner, stakeholder, priority and age.
- Surface high-value unused holds, late release, conflicts and low conversion with direct action links.
- Hold creation and release shall be atomic, reason-coded and approval-controlled by value, quantity, type and timing; overlapping holds require explicit priority resolution.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 02 - Hold Type Master

Purpose: Configure reusable hold categories and their default behavior.

- Maintain VIP, Sponsor, Artist, Media, Corporate, Internal and custom types with code, color and description.
- Set default duration, priority, approval, reminder, maximum quantity, permitted channels and auto-release policy.
- Map type to stakeholder classes, contract requirements, reporting dimensions and permitted roles.
- Hold creation and release shall be atomic, reason-coded and approval-controlled by value, quantity, type and timing; overlapping holds require explicit priority resolution.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 03 - Hold Pool Creation

Purpose: Select and create an explicit pool of held seats or capacity.

- Choose inventory by map, section, row, seat list, zone quantity or uploaded allocation with current-state validation.
- Capture event/performance, type, owner, stakeholder, value, purpose, effective time, expiry and notes.
- Preview available, already held, reserved, sold, blocked and inaccessible inventory before submission.
- Hold creation and release shall be atomic, reason-coded and approval-controlled by value, quantity, type and timing; overlapping holds require explicit priority resolution.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 04 - Hold Rule Assignment

Purpose: Automatically apply approved hold policies to eligible performances.

- Build conditions by tenant, venue, event, product, date, performance, section, category, stakeholder and channel.
- Define seat set or quantity, type, owner, priority, duration, release rule and exception behavior.
- Simulate upcoming performances, detect rule collisions and version every activated rule.
- Hold creation and release shall be atomic, reason-coded and approval-controlled by value, quantity, type and timing; overlapping holds require explicit priority resolution.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 05 - Expiration Rules

Purpose: Control when held inventory must be reviewed or released.

- Set fixed timestamp, time-before-performance, contract milestone, payment condition or manual approval expiry.
- Configure reminder sequence, grace period, escalation, extension eligibility and maximum extension.
- Respect venue timezone and daylight-saving behavior and preserve the rule/version applied to each hold.
- Hold creation and release shall be atomic, reason-coded and approval-controlled by value, quantity, type and timing; overlapping holds require explicit priority resolution.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 06 - Automatic Hold Release

Purpose: Release eligible held inventory reliably and in controlled batches.

- Schedule releases by type, rule, event, performance, stakeholder, age and remaining time.
- Preview seats, price bands, channel availability, sold-out status and exceptions before execution.
- Use idempotent processing, partial-failure handling, retry, reconciliation and stakeholder notification.
- Hold creation and release shall be atomic, reason-coded and approval-controlled by value, quantity, type and timing; overlapping holds require explicit priority resolution.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 07 - Release, Convert & Reassign

Purpose: Allow authorized users to change the outcome of an active hold.

- Release all or selected inventory, convert it to reservation/order, reassign owner/stakeholder or extend expiry.
- Revalidate state, price, eligibility, payment or contract condition before committing the action.
- Require reason, supporting notes and approval when value, timing or policy threshold is exceeded.
- Hold creation and release shall be atomic, reason-coded and approval-controlled by value, quantity, type and timing; overlapping holds require explicit priority resolution.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 08 - Hold Approval Workflow

Purpose: Route hold requests and sensitive changes to the correct approvers.

- Configure requester, venue operations, ticketing, commercial, finance and executive stages by policy.

- Display request details, seats, value, expiry, conflicts, supporting contract and previous decisions.
- Support approve, reject, request information, delegate and escalate with comments and SLA.
- Hold creation and release shall be atomic, reason-coded and approval-controlled by value, quantity, type and timing; overlapping holds require explicit priority resolution.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 09 - Priority & Conflict Resolution

Purpose: Resolve overlapping or competing hold requests predictably.

- Maintain priority matrix by hold type, event, stakeholder, contract, quantity, value and time before performance.
- List exact conflicting seats and compare current and proposed owner, purpose, priority and expiry.
- Allow automatic winner, supervisor decision, alternative seat suggestion or split allocation with recorded reason.
- Hold creation and release shall be atomic, reason-coded and approval-controlled by value, quantity, type and timing; overlapping holds require explicit priority resolution.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 10 - Hold Utilization & Audit

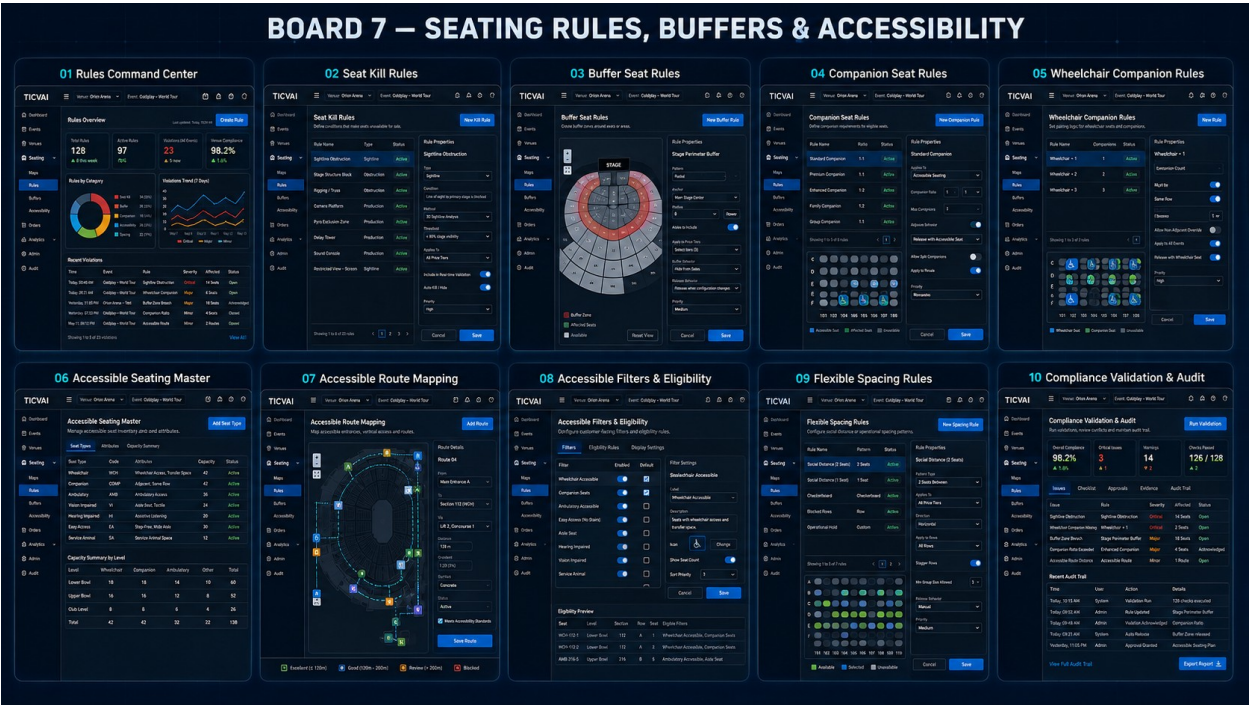
Purpose: Measure whether held inventory creates value or suppresses sale unnecessarily.

- Report held-to-sold conversion, released and sold, expired unused, average age, value and revenue opportunity.
- Compare hold type, owner, stakeholder, event, section and release timing over time.
- Provide immutable creation, approval, extension, conversion, release, conflict and override history with secured export.
- Hold creation and release shall be atomic, reason-coded and approval-controlled by value, quantity, type and timing; overlapping holds require explicit priority resolution.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Board 7 - Seating Rules, Buffers & Accessibility

Figure 7. High-definition configuration board with all 10 screens.



Visual reference: information architecture and configuration coverage; detailed production behavior is defined in the following scope.

Board 7 Scope of Work

Purpose. Configure seat-kill, buffer, companion and accessibility rules while preserving capacity, safety, dignity and an inclusive selection experience.

Source coverage. Matrix sections 21.7 - Seating Rules & Buffer and 21.8 - Accessible Seating

Operating workflow. Define rule and applicability -> select geometry or conditions -> simulate inventory effect -> validate accessibility and circulation -> approve and activate -> enforce during availability/selection -> monitor violations and audit.

Mandatory integrations. Venue Maps, Layouts, Seat Inventory, Event Production, Resource and Maintenance, Customer Accessibility Preferences, Selection Channels, Access Control, Approval, Audit and Compliance Reporting.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Rules Command Center

Purpose: Monitor seating-rule coverage, violations and compliance across venues.

- Show total and active rules, affected seats, open violations, capacity impact and venue compliance score.
- Filter by rule category, tenant, venue, event, layout, effective date, owner and severity.
- Surface missing accessible mappings, conflicting buffers and production changes requiring revalidation.
- Accessibility inventory and companion rules shall be policy-driven and auditable; overrides require authority and reason and must not unlawfully reduce accessible availability.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 02 - Seat Kill Rules

Purpose: Remove seats from sale based on permanent or conditional unusability.

- Configure sightline, stage, production, obstruction, equipment, safety, maintenance and venue-defined kill reasons.
- Apply by seat list, section, row, polygon, obstruction distance, event type or layout condition.
- Preview affected capacity, sold/reserved inventory and alternative seating before activation.
- Accessibility inventory and companion rules shall be policy-driven and auditable; overrides require authority and reason and must not unlawfully reduce accessible availability.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 03 - Buffer Seat Rules

Purpose: Create controlled unsold spacing around seats, zones or production objects.

- Define left/right, radial, row, perimeter, alternating or custom buffer patterns and minimum distance.
- Apply dynamically around selected seats, wheelchair spaces, camera positions, stages or restricted zones.
- Configure release behavior, priority, interaction with holds and treatment when a buffer conflicts with a sale.
- Accessibility inventory and companion rules shall be policy-driven and auditable; overrides require authority and reason and must not unlawfully reduce accessible availability.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 04 - Companion Seat Rules

Purpose: Keep appropriate companion inventory linked to designated seats.

- Configure eligible seat types, adjacency, ratio, maximum companions and same-row or nearby rules.
- Control whether companions must be selected together and when unneeded companion seats may be released.
- Support family, premium, group and venue-defined companion policies without overriding wheelchair-specific rules.
- Accessibility inventory and companion rules shall be policy-driven and auditable; overrides require authority and reason and must not unlawfully reduce accessible availability.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 05 - Wheelchair Companion Rules

Purpose: Manage wheelchair-space and companion-seat pairing.

- Define one-to-one or configurable ratios, adjacency, transfer-seat eligibility and alternative companion locations.
- Require wheelchair space selection before companion inventory where policy permits and explain the rule clearly.
- Control release to general sale only through approved timing, evidence and accessible-demand safeguards.
- Accessibility inventory and companion rules shall be policy-driven and auditable; overrides require authority and reason and must not unlawfully reduce accessible availability.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 06 - Accessible Seating Master

Purpose: Maintain a standard catalog of accessible seat and space attributes.

- Configure wheelchair, companion, ambulatory, transfer, hearing, vision, service-animal and other supported types.
- Store dimensions, capacity treatment, route requirements, amenities, assisted access and display labels.
- Map each type to venue inventory, ticket products, customer-facing filters and compliance reporting.
- Accessibility inventory and companion rules shall be policy-driven and auditable; overrides require authority and reason and must not unlawfully reduce accessible availability.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 07 - Accessible Route Mapping

Purpose: Identify accessible journeys from entry to seats and facilities.

- Create routes through accessible entrances, lifts, ramps, concourses, restrooms and designated seating.
- Store distance, slope, level changes, width, surface, assistance requirement and temporary closure status.
- Validate route continuity and show impact when a layout, facility or access point becomes unavailable.
- Accessibility inventory and companion rules shall be policy-driven and auditable; overrides require authority and reason and must not unlawfully reduce accessible availability.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 08 - Accessible Filters & Eligibility

Purpose: Configure inclusive discovery without exposing sensitive guest information.

- Define filters and labels for wheelchair, companion, aisle, step-free, transfer, hearing, vision and service-animal needs.
- Configure eligibility attestation, documentation policy where lawful, assisted-sale path and privacy restrictions.
- Preview the B2C, mobile, POS and call-center experience and ensure equivalent inventory visibility.
- Accessibility inventory and companion rules shall be policy-driven and auditable; overrides require authority and reason and must not unlawfully reduce accessible availability.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 09 - Flexible Spacing Rules

Purpose: Support temporary public-health or operational spacing requirements.

- Configure distance, seat count, checkerboard, blocked-row, household-group and custom patterns.
- Apply by event, venue, zone, seat type or time window and preview capacity and revenue impact.
- Coordinate with group seating, accessible inventory, holds and sold seats and prevent retroactive unsafe changes.
- Accessibility inventory and companion rules shall be policy-driven and auditable; overrides require authority and reason and must not unlawfully reduce accessible availability.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 10 - Compliance Validation & Audit

Purpose: Demonstrate that seating rules and accessibility configuration are complete and enforced.

- Run validations for accessible quantities, companion ratios, route continuity, conflicts, labels and customer-channel parity.
- Provide issue severity, affected inventory, rule reference, owner, deadline, remediation and evidence.
- Retain configuration, approval, enforcement, release and override history with controlled compliance export.
- Accessibility inventory and companion rules shall be policy-driven and auditable; overrides require authority and reason and must not unlawfully reduce accessible availability.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Board 8 - Group Reservations & Bulk Allocation

Figure 8. High-definition configuration board with all 10 screens.



Visual reference: information architecture and configuration coverage; detailed production behavior is defined in the following scope.

Board 8 Scope of Work

Purpose. Manage corporate, school, tour and other group seat requests from inquiry through bulk allocation, participant assignment, payment and final audit.

Source coverage. Matrix section 21.9 - Group Reservations & Blocks

Operating workflow. Receive group request -> classify and qualify -> search best-fit availability -> create hold/allocation -> quote and approve -> collect deposit/payment -> assign roster -> modify or release -> confirm tickets -> report and audit.

Mandatory integrations. CRM/B2B Accounts, Events, Seat Inventory, Holds, Pricing, Promotions, Quotes, Contracts, Payments, Orders, Ticketing, Communications, Approval and Reporting.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Group Reservation Center

Purpose: Provide a pipeline view of all group inquiries and reservations.

- Show new requests, quoted, held, awaiting deposit, confirmed, expired, cancelled and completed groups.
- Display seats, value, conversion, deadline, outstanding payment and inventory-release risk.
- Filter by group type, account, event, venue, owner, stage, date and priority with saved views.
- Bulk allocation shall use the same real-time inventory and hold controls as individual sale; discounts, deposits, release dates and contract approvals follow shared finance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 02 - Group Type Configuration

Purpose: Define reusable rules for corporate, school, tour and custom groups.

- Maintain minimum/maximum size, default hold days, deposit, payment schedule, release timing and cancellation policy.
- Configure auto-allocation, contiguity, split allowance, companion/adult ratio, contract and approval requirements.
- Map type to CRM account categories, pricing/discount policy, communication templates and reports.
- Bulk allocation shall use the same real-time inventory and hold controls as individual sale; discounts, deposits, release dates and contract approvals follow shared finance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 03 - Group Request Intake

Purpose: Capture a complete group seating requirement from any authorized channel.

- Record account/contact, event, performance, expected size, age mix, budget, zone, accessibility and seating preferences.
- Capture arrival, transport, guide, invoice, tax, purchase-order, contract and special service information as applicable.
- Detect duplicates, validate capacity/time and support draft, submit and assignment to a sales owner.
- Bulk allocation shall use the same real-time inventory and hold controls as individual sale; discounts, deposits, release dates and contract approvals follow shared finance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 04 - Availability & Best-Fit Search

Purpose: Find the strongest seating options for a group request.

- Search live inventory by group size, contiguous requirement, split limit, price, zone, view and accessibility.
- Rank sections or seat groups by fit score, average price, distance, fragmentation and operational constraints.
- Allow comparison, map preview, alternative performance and immediate controlled hold creation.
- Bulk allocation shall use the same real-time inventory and hold controls as individual sale; discounts, deposits, release dates and contract approvals follow shared finance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 05 - Bulk Seat Allocation

Purpose: Allocate large quantities without compromising seat-state integrity.

- Select section, row, seat range, polygon or recommended group and validate all inventory atomically.
- Configure contiguous groups, aisle breaks, split sections, wheelchair/companion pairings and staff seats.
- Create the linked hold/reservation with value, expiry, owner and exact allocated-seat list.
- Bulk allocation shall use the same real-time inventory and hold controls as individual sale; discounts, deposits, release dates and contract approvals follow shared finance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 06 - Roster & Participant Assignment

Purpose: Assign named or placeholder participants to allocated seats.

- Import or enter participant name, type, age band, accessibility need, ticket type and external reference.
- Assign manually or automatically while preserving family, class, coach, guide and accessibility groupings.
- Show unassigned, duplicate, ineligible and missing-information issues and synchronize final assignments to tickets.
- Bulk allocation shall use the same real-time inventory and hold controls as individual sale; discounts, deposits, release dates and contract approvals follow shared finance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 07 - Quote, Deposit & Payment

Purpose: Manage the commercial proposal and collection schedule for the group.

- Build quote lines for seats, fees, add-ons, discounts, tax, complimentary inventory and optional services.
- Apply shared discount authority, price validity, deposit percentage/fixed amount, milestones and payment methods.
- Track sent, viewed, accepted, partially paid, overdue and paid status and release inventory according to policy.
- Bulk allocation shall use the same real-time inventory and hold controls as individual sale; discounts, deposits, release dates and contract approvals follow shared finance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 08 - Modify, Release & Cancel

Purpose: Control changes to confirmed or pending group reservations.

- Increase, reduce, reseat, change performance, release selected seats or cancel all/part of the group.
- Preview price, fee, deposit, refund, cancellation charge, participant and ticket impact before confirmation.
- Reacquire or release inventory atomically and route threshold exceptions for approval.
- Bulk allocation shall use the same real-time inventory and hold controls as individual sale; discounts, deposits, release dates and contract approvals follow shared finance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 09 - Contracts & Approval Workflow

Purpose: Manage documents and approval stages required for group business.

- Attach or generate proposal, contract, purchase order, roster, waiver requirement and supporting documents.
- Route pricing, credit, deposit, allocation, complimentary seats and cancellation exceptions to configured approvers.
- Track document version, e-signature, approval SLA, comments and final evidence without duplicating the document service.
- Bulk allocation shall use the same real-time inventory and hold controls as individual sale; discounts, deposits, release dates and contract approvals follow shared finance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 10 - Group Reporting & Audit

Purpose: Measure group-sales performance and preserve transaction accountability.

- Report pipeline, conversion, seats, revenue, yield, discount, outstanding payment, cancellation and utilization.
- Compare corporate, school, tour, account, sales owner, event, section and lead source.
- Audit inquiry, allocation, hold, quote, approval, payment, roster, ticket, modification and release events.
- Bulk allocation shall use the same real-time inventory and hold controls as individual sale; discounts, deposits, release dates and contract approvals follow shared finance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Board 9 - AI Seat Recommendations & Optimization

Figure 9. High-definition configuration board with all 10 screens.



Visual reference: information architecture and configuration coverage; detailed production behavior is defined in the following scope.

Board 9 Scope of Work

Purpose. Provide explainable best-seat, value, proximity, family, accessibility, upgrade and alternative recommendations using live governed inventory.

Source coverage. Matrix section 21.10 - Seat Recommendations

Operating workflow. Capture objective and constraints -> retrieve live eligible inventory -> score candidates -> apply adjacency/accessibility/pricing rules -> rank and explain -> lock selected option -> collect outcome/feedback -> monitor model performance.

Mandatory integrations. Seat Inventory, Venue Geometry, Layout/Focal Point, Pricing, Customer Preferences, Accessibility, Cart, Orders, Membership/Loyalty, Analytics, AI Model Registry, Approval and Audit.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Recommendation Command Center

Purpose: Monitor recommendation demand, quality, conversion and risk.

- Show recommendations served, accepted, locked, purchased, revenue uplift, fallback and no-result rates.
- Compare use case, channel, venue, performance, model/version, guest type and time period.
- Surface data gaps, drift, declining acceptance, accessibility mismatch and model/service incidents.
- Recommendations must be explainable, confidence-scored and subordinate to availability, price, lock, accessibility, eligibility and consent rules; models cannot write inventory state directly.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 02 - Best Seat Recommendations

Purpose: Rank the strongest overall options for the requested quantity.

- Score view, distance, orientation, row, aisle, obstruction, amenities, demand and customer preference.
- Return valid seat groups with match score, total price, view summary, rationale and trade-offs.
- Support rule-based fallback when AI is unavailable or confidence falls below the configured threshold.
- Recommendations must be explainable, confidence-scored and subordinate to availability, price, lock, accessibility, eligibility and consent rules; models cannot write inventory state directly.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 03 - Best Value Recommendations

Purpose: Balance quality and price to identify strong value options.

- Calculate value from price, view, distance, category, demand, historical preference and comparable inventory.
- Present a price-versus-quality matrix and explain why each option represents better value.
- Respect minimum/maximum budget, promotion eligibility, fee transparency and price-lock policy.
- Recommendations must be explainable, confidence-scored and subordinate to availability, price, lock, accessibility, eligibility and consent rules; models cannot write inventory state directly.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 04 - Closest-to-Stage Recommendations

Purpose: Rank seats by meaningful proximity to the event focal point.

- Use configured stage/field/focal point, seat coordinates, orientation, level and route rather than row number alone.
- Show distance, section, row, elevation, obstruction and route information for each option.
- Support multiple focal points and event-specific stage layouts with the correct layout version.
- Recommendations must be explainable, confidence-scored and subordinate to availability, price, lock, accessibility, eligibility and consent rules; models cannot write inventory state directly.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 05 - Family Seating Recommendations

Purpose: Find safe and practical seat groups for families.

- Prioritize contiguity, aisle preference, child-friendly areas, short routes and proximity to approved facilities.
- Consider adult/child ratio, age policy, stroller/service needs, family membership and budget.
- Explain splits or compromises and never separate minors contrary to configured policy.
- Recommendations must be explainable, confidence-scored and subordinate to availability, price, lock, accessibility, eligibility and consent rules; models cannot write inventory state directly.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 06 - Accessibility Recommendations

Purpose: Match stated access needs to verified accessible inventory and routes.

- Use wheelchair, companion, transfer, aisle, step-free, hearing, vision and service-animal attributes.
- Show companion pairing, accessible route, distance to entrance/facilities and any assistance requirement.
- Protect sensitive preference data, allow human-assisted review and prevent inappropriate release of protected inventory.
- Recommendations must be explainable, confidence-scored and subordinate to availability, price, lock, accessibility, eligibility and consent rules; models cannot write inventory state directly.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 07 - Seat Upgrade Recommendations

Purpose: Offer an eligible higher-value alternative before or after purchase.

- Compare current and candidate seat on view, distance, level, amenities, accessibility and price difference.
- Apply membership, loyalty, promotion, exchange, refund, fee and event cutoff rules.
- Show expected value and acquire new inventory before releasing the original seat through a safe exchange flow.
- Recommendations must be explainable, confidence-scored and subordinate to availability, price, lock, accessibility, eligibility and consent rules; models cannot write inventory state directly.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 08 - Alternatives & Reseating

Purpose: Provide comparable options when selected or issued seats become unavailable.

- Rank same-price, better-price, better-view, same-section, accessible and adjacent alternatives.
- Support operational reseating for layout changes, maintenance, production blocks and customer requests.
- Explain impact, require approval/acceptance where applicable and preserve original-to-new seat lineage.
- Recommendations must be explainable, confidence-scored and subordinate to availability, price, lock, accessibility, eligibility and consent rules; models cannot write inventory state directly.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 09 - Scoring Rules & Model Governance

Purpose: Configure recommendation objectives, constraints and AI controls.

- Set weighted factors, hard constraints, tie-breakers, minimum score, fallback, audience and use-case priority.
- Manage model/version, training/evaluation metadata, approval, effective dates and rollback.
- Run test personas, backtests, bias/accessibility checks and scenario comparison before publication.
- Recommendations must be explainable, confidence-scored and subordinate to availability, price, lock, accessibility, eligibility and consent rules; models cannot write inventory state directly.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 10 - Performance, Feedback & Audit

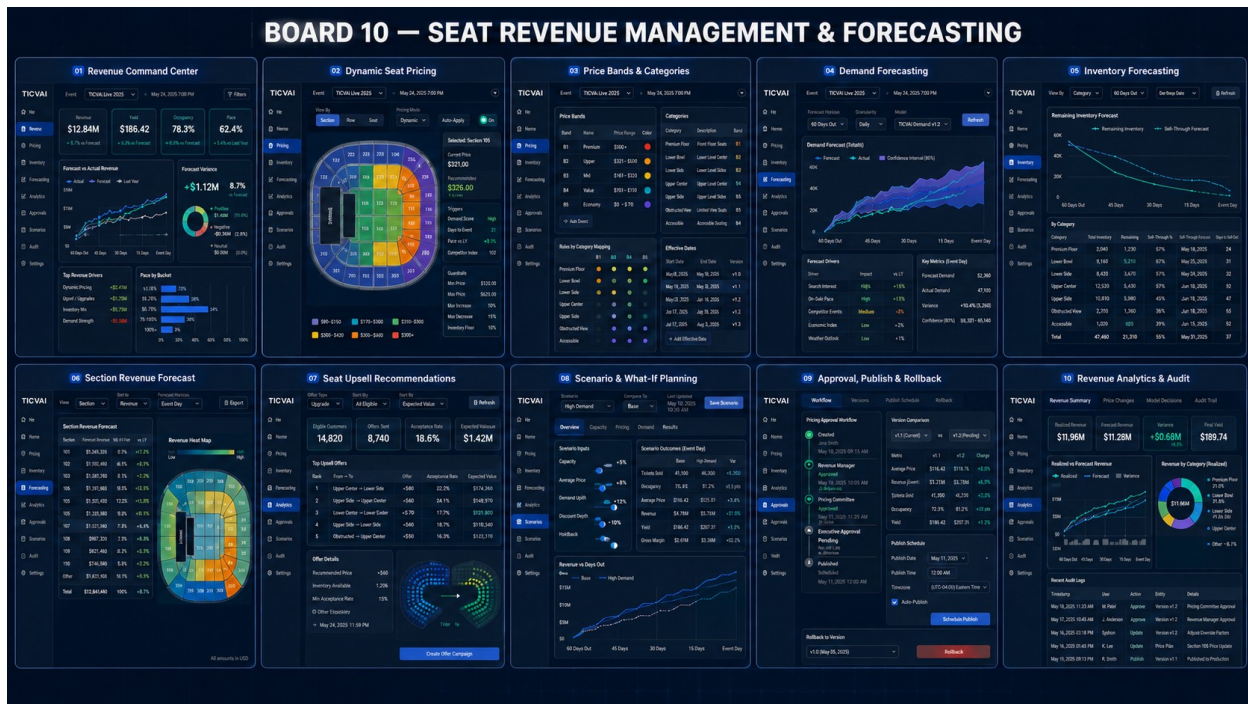
Purpose: Measure realized recommendation quality and accountability.

- Report acceptance, conversion, revenue, upgrade, abandonment, no-result and downstream satisfaction.
- Capture accept/reject, reason, user feedback, actual purchase and post-event outcome for learning.
- Monitor drift and disparity and retain inputs, candidates, scores, explanation, model/version and human decisions.
- Recommendations must be explainable, confidence-scored and subordinate to availability, price, lock, accessibility, eligibility and consent rules; models cannot write inventory state directly.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Board 10 - Seat Revenue Management & Forecasting

Figure 10. High-definition configuration board with all 10 screens.



Visual reference: information architecture and configuration coverage; detailed production behavior is defined in the following scope.

Board 10 Scope of Work

Purpose. Optimize seat yield through governed dynamic pricing, demand and inventory forecasting, section-level revenue planning and eligible upsell recommendations.

Source coverage. Matrix section 21.11 - Seat Revenue Management & Forecasting

Operating workflow. Ingest sales pace, inventory and demand signals -> forecast demand and remaining inventory -> define or recommend price/upsell action -> simulate guardrails and impact -> review and approve -> publish -> measure realized outcome and audit.

Mandatory integrations. Seat Inventory, Events, Product Catalog, Pricing, Promotions, Orders, Payments, CRM/Segmentation, Membership/Loyalty, Finance, Analytics, AI Model Registry, Approval and Audit.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Revenue Command Center

Purpose: Provide a live executive and revenue-management view of seat performance.

- Show revenue, yield, occupancy, sales pace, average ticket price, remaining inventory and forecast variance.
- Analyze by tenant, venue, event, performance, section, category, price band, channel and time to event.
- Surface underperforming sections, demand spikes, inventory imbalance and pricing opportunities with drill-down.
- Pricing changes require explainable inputs, min/max guardrails, effective dates, approval, versioning and rollback; AI may recommend but cannot publish outside delegated authority.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 02 - Dynamic Seat Pricing

Purpose: Configure rules that adjust assigned-seat prices within controlled boundaries.

- Apply rules by section, row, seat, category, event, performance, sales pace, demand, remaining inventory and time to event.
- Set base price, adjustment type, floor, ceiling, step, maximum frequency, freeze window and competitor/manual inputs where approved.
- Preview affected inventory, customer display, active carts, taxes/fees, promotions and forecast impact before activation.
- Pricing changes require explainable inputs, min/max guardrails, effective dates, approval, versioning and rollback; AI may recommend but cannot publish outside delegated authority.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 03 - Price Bands & Categories

Purpose: Maintain the commercial hierarchy applied to seat inventory.

- Configure price band, seat category, display label, color, currency, channel, customer segment and effective period.
- Map bands to venue sections, rows or seats and support event/performance overrides with inheritance.
- Detect overlapping dates, unmapped inventory, invalid currency, missing products and conflicting priority.

- Pricing changes require explainable inputs, min/max guardrails, effective dates, approval, versioning and rollback; AI may recommend but cannot publish outside delegated authority.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 04 - Demand Forecasting

Purpose: Forecast seat demand over the full booking horizon.

- Use historical events, sales pace, day/time, artist/team, seasonality, campaigns, price, inventory and external factors where approved.
- Display forecast, actual, confidence interval, key drivers and variance by section/category and days to event.
- Support model/version selection, backtesting, manual adjustment with reason and accuracy monitoring.
- Pricing changes require explainable inputs, min/max guardrails, effective dates, approval, versioning and rollback; AI may recommend but cannot publish outside delegated authority.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 05 - Inventory Forecasting

Purpose: Predict remaining inventory and sell-through by event day.

- Forecast seats remaining, sell-through, sold-out probability and over/under supply by section and category.
- Account for active locks, holds, scheduled releases, blocks, expected cancellations and group allocations.
- Highlight categories likely to sell out early or remain unsold and link to recommended actions.
- Pricing changes require explainable inputs, min/max guardrails, effective dates, approval, versioning and rollback; AI may recommend but cannot publish outside delegated authority.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 06 - Section Revenue Forecast

Purpose: Plan expected revenue at the venue-section level.

- Show capacity, sold, held, available, average price, forecast occupancy, forecast revenue and variance.
- Provide map heat view and ranking by revenue opportunity, yield gap and risk.
- Reconcile forecast totals with performance and finance dimensions and preserve scenario assumptions.
- Pricing changes require explainable inputs, min/max guardrails, effective dates, approval, versioning and rollback; AI may recommend but cannot publish outside delegated authority.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 07 - Seat Upsell Recommendations

Purpose: Configure revenue-positive seat offers that remain fair and eligible.

- Rank upgrade pairs by view improvement, distance, amenities, price delta, availability and customer eligibility.
- Set channels, offer window, frequency, minimum improvement, margin, membership/loyalty benefit and exclusion rules.
- Track offer, acceptance, incremental revenue, original-seat resale and guest outcome.

- Pricing changes require explainable inputs, min/max guardrails, effective dates, approval, versioning and rollback; AI may recommend but cannot publish outside delegated authority.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 08 - Scenario & What-If Planning

Purpose: Compare alternative price, demand and inventory decisions before publication.

- Adjust capacity, price, discount, holdback, release timing, demand uplift and sales pace assumptions.
- Calculate seats sold, occupancy, revenue, yield, gross margin, sell-out timing and downside risk.
- Save, compare, comment, share and approve scenarios without changing live prices or inventory.
- Pricing changes require explainable inputs, min/max guardrails, effective dates, approval, versioning and rollback; AI may recommend but cannot publish outside delegated authority.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 09 - Approval, Publish & Rollback

Purpose: Govern revenue-management changes from draft to production.

- Route pricing and model changes by magnitude, value, event, venue and responsible business unit.
- Compare current/proposed prices, forecast outcomes, affected inventory, channels and policy checks.
- Schedule publication, verify downstream synchronization and support emergency pause or rollback to an approved version.
- Pricing changes require explainable inputs, min/max guardrails, effective dates, approval, versioning and rollback; AI may recommend but cannot publish outside delegated authority.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 10 - Revenue Analytics & Audit

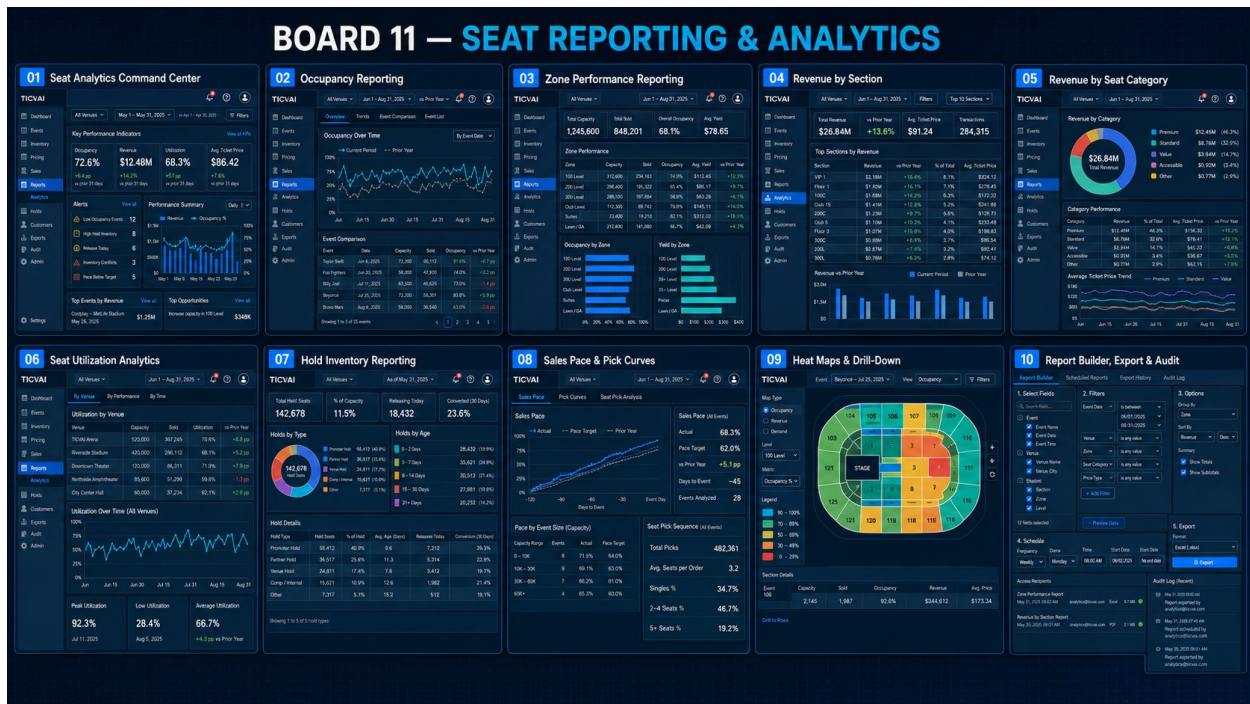
Purpose: Measure actual financial outcomes and preserve decision evidence.

- Compare realized versus forecast revenue, occupancy, yield, average price and sell-through by section/category.
- Attribute change to price actions, holds/releases, promotions, demand shifts, channels and model recommendations.
- Retain rule, input data version, model/version, recommendation, approval, publication, override and rollback history.
- Pricing changes require explainable inputs, min/max guardrails, effective dates, approval, versioning and rollback; AI may recommend but cannot publish outside delegated authority.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Board 11 - Seat Reporting & Analytics

Figure 11. High-definition configuration board with all 10 screens.



Visual reference: information architecture and configuration coverage; detailed production behavior is defined in the following scope.

Board 11 Scope of Work

Purpose. Deliver trusted occupancy, zone, revenue, utilization, hold and seat-pick analytics with governed drill-down, scheduling and export.

Source coverage. Matrix section 21.12 - Seat Reporting & Analytics

Operating workflow. Select governed dataset -> filter and aggregate -> visualize occupancy/revenue/utilization -> drill to section, row or seat -> schedule/share controlled report -> reconcile metric definition -> retain export and access audit.

Mandatory integrations. Seat Inventory, Events, Orders, Payments/Finance, Holds, Reservations, Pricing, Access Control, Data Warehouse/Lakehouse, BI, Notification, Security and Audit.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Seat Analytics Command Center

Purpose: Provide a consolidated view of seat commercial and operational performance.

- Show occupancy, revenue, utilization, average ticket price, held inventory, release yield and forecast variance.
- Present trend, benchmark, target and prior-period comparison by tenant, venue and event.
- Surface low occupancy, excessive holds, inventory conflict, price variance and data-quality alerts.
- Metrics shall use versioned definitions and authoritative sources; row/seat and guest-level drill-down, exports and subscriptions follow RBAC/PBAC and data-retention rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 02 - Occupancy Reporting

Purpose: Report sold and attended occupancy against configured capacity.

- Calculate sold, issued, scanned, no-show, blocked, available and effective capacity with clear metric definitions.
- Analyze by event, performance, venue, section, category, channel and time.
- Compare events and periods and drill to approved seat-level evidence.
- Metrics shall use versioned definitions and authoritative sources; row/seat and guest-level drill-down, exports and subscriptions follow RBAC/PBAC and data-retention rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 03 - Zone Performance Reporting

Purpose: Compare capacity, demand and yield across venue zones.

- Show zone capacity, sold, occupancy, average price, revenue, utilization, holds and release conversion.
- Rank zones and compare target, forecast, prior period and venue benchmark.
- Drill from zone to section, row and seat while preserving the selected metric and filters.
- Metrics shall use versioned definitions and authoritative sources; row/seat and guest-level drill-down, exports and subscriptions follow RBAC/PBAC and data-retention rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 04 - Revenue by Section

Purpose: Analyze financial contribution from each seating section.

- Report gross/net revenue, tax, fees, discounts, refunds, average ticket price and percentage of performance total.
- Compare section revenue, yield, occupancy, price band and prior-year/forecast variance.
- Link summarized values to governed finance/order detail and reconciliation status.
- Metrics shall use versioned definitions and authoritative sources; row/seat and guest-level drill-down, exports and subscriptions follow RBAC/PBAC and data-retention rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 05 - Revenue by Seat Category

Purpose: Measure performance of Premium, Standard, Value, Accessible and configured categories.

- Show revenue mix, seats sold, average price, discount, refund, yield and sell-through by category.
- Compare venue, event, performance, channel, customer segment and days to event.
- Preserve accessible-category privacy and prevent misleading comparison caused by protected inventory rules.
- Metrics shall use versioned definitions and authoritative sources; row/seat and guest-level drill-down, exports and subscriptions follow RBAC/PBAC and data-retention rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 06 - Seat Utilization Analytics

Purpose: Measure how effectively physical and sellable seating capacity is used.

- Calculate booked, sold, scanned, occupied estimate, idle and unavailable hours or performances as applicable.
- Analyze utilization by venue, map, section, seat type, day, time and event category.
- Identify permanently underused areas, recurring blocks, maintenance patterns and lost-sale opportunities.
- Metrics shall use versioned definitions and authoritative sources; row/seat and guest-level drill-down, exports and subscriptions follow RBAC/PBAC and data-retention rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 07 - Hold Inventory Reporting

Purpose: Report the size, age and outcome of seat holds.

- Show held quantity/value, capacity percentage, active age, upcoming release, conversion and expired unused.
- Compare VIP, Sponsor, Artist, Media, Corporate, Internal and custom types by owner/stakeholder.
- Measure released-to-sold outcomes and potential revenue suppressed by late or unused holds.
- Metrics shall use versioned definitions and authoritative sources; row/seat and guest-level drill-down, exports and subscriptions follow RBAC/PBAC and data-retention rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 08 - Sales Pace & Pick Curves

Purpose: Analyze when seats sell and which inventory customers select first.

- Plot cumulative sales by days/hours to event against target, forecast and comparable events.

- Show seat-pick sequence by section, row, price, view and channel and identify preferred inventory patterns.
- Support cohort comparison and expose sufficient sample and confidence information for decisions.
- Metrics shall use versioned definitions and authoritative sources; row/seat and guest-level drill-down, exports and subscriptions follow RBAC/PBAC and data-retention rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 09 - Heat Maps & Drill-Down

Purpose: Visualize spatial performance directly on the approved seat map.

- Display occupancy, revenue, average price, demand, availability, holds, scan and utilization heat maps.
- Switch level, section, row and seat granularity and synchronize filters with tables and charts.
- Open the supporting transactions where authorized and show metric definition, source and refresh time.
- Metrics shall use versioned definitions and authoritative sources; row/seat and guest-level drill-down, exports and subscriptions follow RBAC/PBAC and data-retention rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 10 - Report Builder, Export & Audit

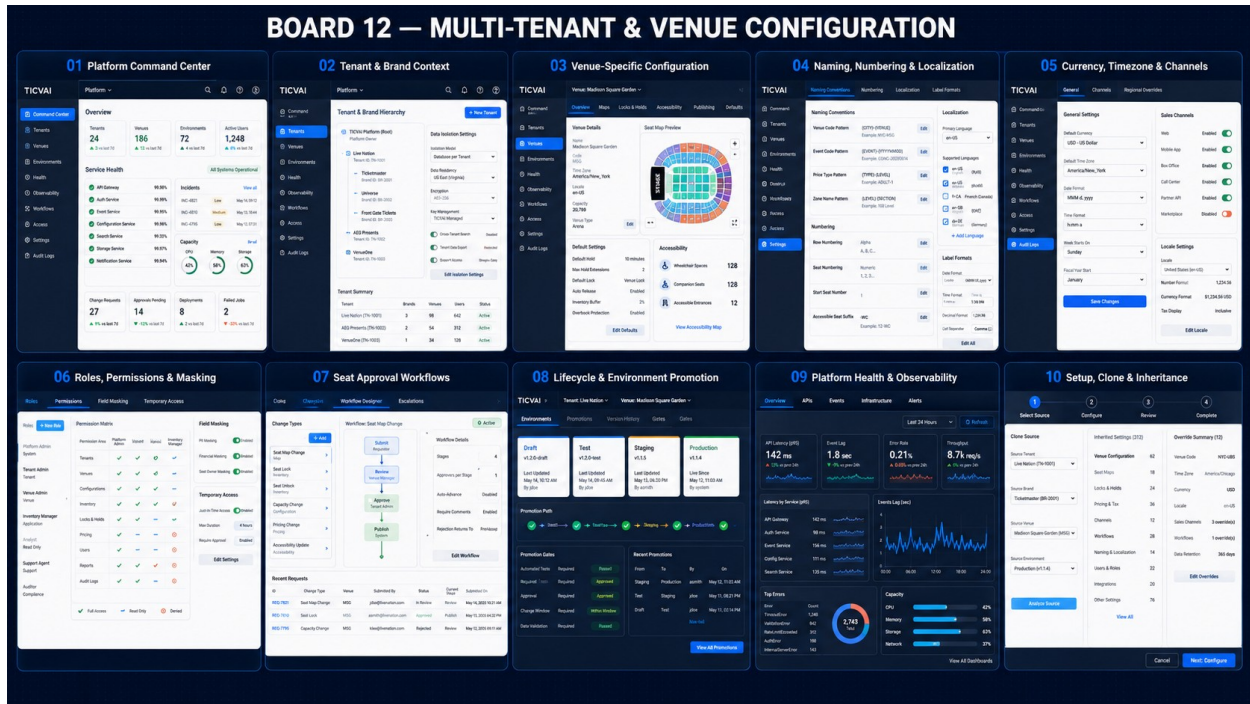
Purpose: Allow authorized users to assemble and distribute controlled seat reports.

- Select approved dimensions, measures, filters, grouping, sorting, chart/map and date context.
- Save personal/team views and schedule email or in-app delivery with recipient and permission validation.
- Export CSV, XLSX or PDF subject to policy and log report definition, recipients, downloads and sensitive access.
- Metrics shall use versioned definitions and authoritative sources; row/seat and guest-level drill-down, exports and subscriptions follow RBAC/PBAC and data-retention rules.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Board 12 - Multi-Tenant & Venue Configuration

Figure 12. High-definition configuration board with all 10 screens.



Visual reference: information architecture and configuration coverage; detailed production behavior is defined in the following scope.

Board 12 Scope of Work

Purpose. Operate one secure seat platform across multiple tenants, brands, regions and venues while supporting local configuration, inheritance and controlled environment promotion.

Source coverage. Matrix requirements 21.13.1 - Multi-Tenant Support and 21.13.2 - Venue-Specific Configuration, expanded with required platform administration

Operating workflow. Create tenant/brand context -> configure venue defaults and locale -> assign permissions and masking -> define approvals -> validate inheritance -> promote through environments -> monitor service health -> audit configuration and access.

Mandatory integrations. Tenant and Identity Services, Venue Master, Configuration Service, Seat Inventory, API Gateway, Event Bus, Secrets/KMS, Approval, Observability, Notification, Deployment and Audit.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Platform Command Center

Purpose: Provide a platform-wide view of tenant, venue and environment readiness.

- Show tenants, brands, venues, environments, active users, pending approvals, deployments and failed jobs.
- Display service status, API health, capacity, incidents and configuration exceptions.
- Restrict cross-tenant visibility to authorized platform roles and preserve every context switch.
- Enforce tenant isolation and environment separation at UI, API, data, cache, event and export layers; venue overrides cannot weaken mandatory platform security or compliance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 02 - Tenant & Brand Context

Purpose: Configure the hierarchy and isolation boundary for seat management.

- Maintain tenant, legal entity, brand, region, venue ownership, status and data-residency classification.
- Configure tenant isolation, encryption context, export restriction, cross-brand sharing and support-access policy.
- Show inherited platform settings and the venues, users, integrations and environments within each context.
- Enforce tenant isolation and environment separation at UI, API, data, cache, event and export layers; venue overrides cannot weaken mandatory platform security or compliance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 03 - Venue-Specific Configuration

Purpose: Set local defaults without creating separate seat systems.

- Configure default map, lock/hold duration, naming, accessibility, sales channels, approvals and publishing policy.
- Map local entrances, devices, price/seat categories, operating calendars and venue-specific exceptions.
- Display which settings inherit from platform/tenant/brand and require justification for permitted overrides.
- Enforce tenant isolation and environment separation at UI, API, data, cache, event and export layers; venue overrides cannot weaken mandatory platform security or compliance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 04 - Naming, Numbering & Localization

Purpose: Control how seat entities and values appear in each venue and language.

- Configure venue/section/row/seat code formats, numbering direction, padding, prefixes and reserved values.
- Maintain translated labels, supported languages, text direction, pluralization and fallback language.
- Validate uniqueness and preview labels on map, ticket, cart, POS, access device and report.
- Enforce tenant isolation and environment separation at UI, API, data, cache, event and export layers; venue overrides cannot weaken mandatory platform security or compliance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 05 - Currency, Timezone & Channels

Purpose: Configure regional display and channel availability.

- Set currency, rounding, date/time/number formats, timezone, week start and daylight-saving behavior.
- Enable web, mobile, POS, box office, call center, B2B, partner API and marketplace channels per venue.
- Define locale defaults and fail-safe behavior when a channel or currency is unavailable.
- Enforce tenant isolation and environment separation at UI, API, data, cache, event and export layers; venue overrides cannot weaken mandatory platform security or compliance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 06 - Roles, Permissions & Masking

Purpose: Enforce least-privilege access to seat administration and operational data.

- Configure RBAC/PBAC for maps, layouts, inventory, locks, holds, pricing, reports, integrations and audit.
- Apply venue/region scope, field masking, temporary/delegated access, segregation of duties and emergency override.
- Preview effective access for a user/context and log sensitive read, export and administrative actions.
- Enforce tenant isolation and environment separation at UI, API, data, cache, event and export layers; venue overrides cannot weaken mandatory platform security or compliance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 07 - Seat Approval Workflows

Purpose: Configure governance paths for high-impact seat changes.

- Define workflow by map/layout, capacity, seat kill, hold, accessibility, pricing, integration and override change type.
- Set stages, approver roles, thresholds, quorum, SLA, escalation, delegation, comments and evidence.
- Prevent requester self-approval where segregation rules apply and preserve decision history.
- Enforce tenant isolation and environment separation at UI, API, data, cache, event and export layers; venue overrides cannot weaken mandatory platform security or compliance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 08 - Lifecycle & Environment Promotion

Purpose: Move approved configuration safely from draft to production.

- Provide Draft, Test, Staging and Production contexts with version, owner, status and dependency checks.
- Promote signed configuration packages through required gates, automated tests and approvals.
- Support comparison, rollback, failed-promotion recovery and prohibition of direct unapproved production edits.
- Enforce tenant isolation and environment separation at UI, API, data, cache, event and export layers; venue overrides cannot weaken mandatory platform security or compliance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 09 - Platform Health & Observability

Purpose: Monitor the technical services that protect seat-state correctness.

- Show API latency/error, event lag, lock service, inventory service, cache, database, queue and webhook health.
- Configure SLOs, thresholds, alerts, correlation, incident ownership and capacity forecasts by tenant/venue.
- Drill from a metric to traces, logs and affected performances without exposing restricted tenant data.
- Enforce tenant isolation and environment separation at UI, API, data, cache, event and export layers; venue overrides cannot weaken mandatory platform security or compliance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 10 - Setup, Clone & Inheritance

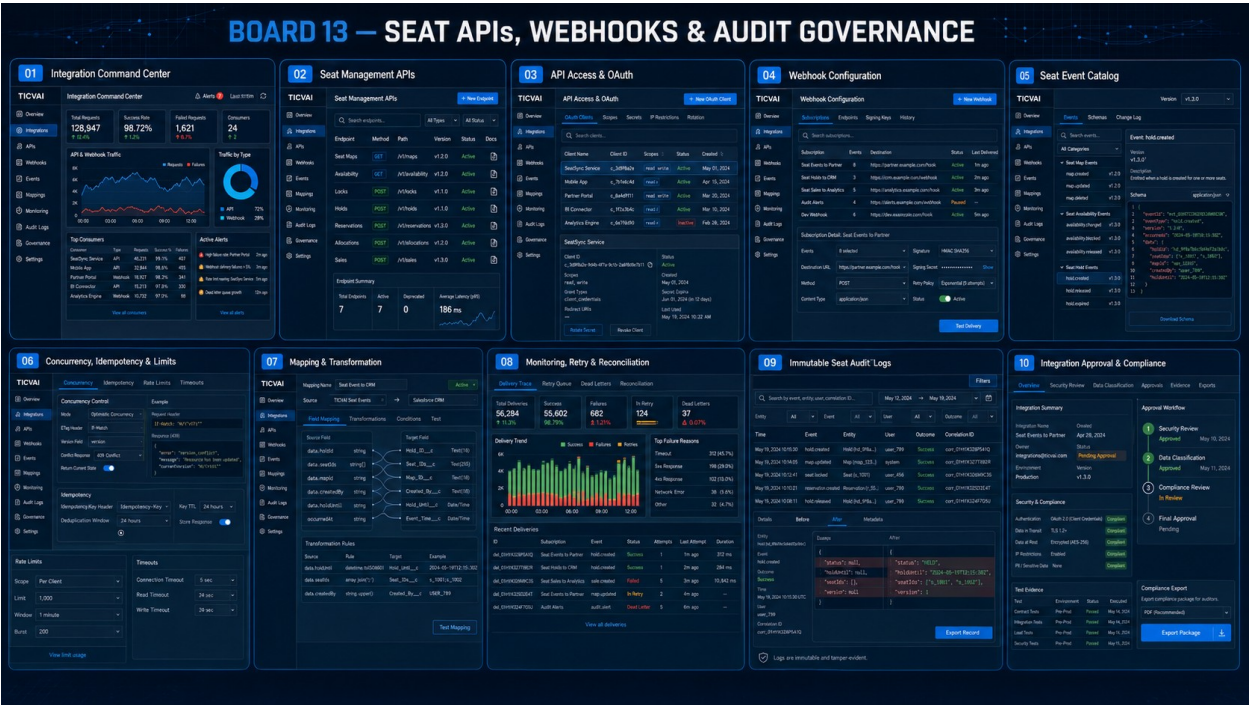
Purpose: Accelerate onboarding while keeping configuration lineage visible.

- Guide setup of tenant, brand, venue, locale, map, categories, locks/holds, channels, workflows and integrations.
- Clone an authorized source venue or template and choose which settings, maps and rules to copy or inherit.
- Show inherited, overridden, missing and conflicting settings and validate completeness before activation.
- Enforce tenant isolation and environment separation at UI, API, data, cache, event and export layers; venue overrides cannot weaken mandatory platform security or compliance policies.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Board 13 - Seat APIs, Webhooks & Audit Governance

Figure 13. High-definition configuration board with all 10 screens.



Visual reference: information architecture and configuration coverage; detailed production behavior is defined in the following scope.

Board 13 Scope of Work

Purpose. Expose secure, versioned seat services and events to approved channels and partners with reliable delivery, reconciliation and tamper-evident governance.

Source coverage. Matrix requirements 21.13.3 - Seat Management APIs, 21.13.4 - Webhooks, 21.13.5 - Seat Audit Logs and 21.13.6 - Approval Workflows

Operating workflow. Register consumer -> authorize scopes -> discover versioned API/event -> map and test -> approve -> activate -> monitor request/delivery -> retry or reconcile failure -> rotate credentials -> review audit and compliance evidence.

Mandatory integrations. API Gateway, Identity/OAuth, Seat Inventory, Venue/Layouts, Cart/Orders, Event Bus, Webhook Delivery, Secrets/KMS, SIEM, Observability, Approval, Developer Portal and Audit Archive.

Board acceptance. All ten screens shall be implemented as connected production features, with responsive states, validation, permissions, approvals where applicable, API/event integration, error handling, monitoring and immutable audit evidence.

Screen 01 - Integration Command Center

Purpose: Monitor all seat API and webhook consumers from one place.

- Show request volume, success, latency, failures, retries, active consumers, versions and security alerts.
- Compare tenant, venue, application, endpoint, event, environment and time period.
- Surface deprecated clients, abnormal traffic, delivery backlog and reconciliation failures with owner and action.
- All writes require authenticated, authorized, idempotent and concurrency-safe requests; events and audit records carry tenant, version and correlation context and are retained under approved policy.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 02 - Seat Management APIs

Purpose: Publish a governed catalog of supported seat endpoints.

- Document APIs for maps/layouts, availability, locks, holds, reservations, allocations, sales and seat history.
- Show method, path, version, scope, request/response schema, errors, examples, limits and deprecation date.
- Provide sandbox testing and prevent production calls until client, scopes and integration approval are active.
- All writes require authenticated, authorized, idempotent and concurrency-safe requests; events and audit records carry tenant, version and correlation context and are retained under approved policy.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 03 - API Access & OAuth

Purpose: Manage machine and delegated access to seat services.

- Register confidential/public clients with tenant, environment, owner, redirect/callback and application purpose.
- Configure OAuth grant, scopes, token lifetime, mTLS/IP restriction, secret/certificate rotation and revocation.
- Display secret metadata without exposing stored secret values and audit credential and scope changes.
- All writes require authenticated, authorized, idempotent and concurrency-safe requests; events and audit records carry tenant, version and correlation context and are retained under approved policy.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 04 - Webhook Configuration

Purpose: Allow approved consumers to subscribe to seat lifecycle events.

- Select events, tenant/venue scope, destination, method, content type, version and filtering conditions.
- Configure HMAC signature, signing-key rotation, timeout, retry, ordering, batching and dead-letter policy.
- Test challenge and sample delivery, verify endpoint ownership and block activation until security checks pass.
- All writes require authenticated, authorized, idempotent and concurrency-safe requests; events and audit records carry tenant, version and correlation context and are retained under approved policy.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 05 - Seat Event Catalog

Purpose: Define the versioned events emitted by the seat platform.

- Catalog map/layout published, availability changed, seat locked/unlocked, held/released, reserved, sold and blocked events.
- Show schema, required tenant/venue/performance/seat identifiers, timestamp, version and correlation metadata.
- Manage backward compatibility, deprecation, sample payload and consumer impact for schema changes.
- All writes require authenticated, authorized, idempotent and concurrency-safe requests; events and audit records carry tenant, version and correlation context and are retained under approved policy.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 06 - Concurrency, Idempotency & Limits

Purpose: Configure technical controls that prevent duplicate or conflicting inventory actions.

- Define expected version/ETag, lock token, conflict response and retry boundary for write operations.
- Require idempotency key, retention window and duplicate-response behavior for supported commands.
- Set rate, burst, connection, read/write timeout and payload limits by client, endpoint and environment.
- All writes require authenticated, authorized, idempotent and concurrency-safe requests; events and audit records carry tenant, version and correlation context and are retained under approved policy.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 07 - Mapping & Transformation

Purpose: Map external partner fields and codes to TICVAI canonical seat entities.

- Configure source-to-target identifiers, status/category code mapping, value transformation and defaults.
- Validate data type, required field, tenant/venue context, referential integrity and unsupported values.
- Provide test payload, expected result, versioning and safe rollback without modifying the canonical model.
- All writes require authenticated, authorized, idempotent and concurrency-safe requests; events and audit records carry tenant, version and correlation context and are retained under approved policy.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 08 - Monitoring, Retry & Reconciliation

Purpose: Operate API and event delivery reliably through failure.

- Trace request or webhook by correlation ID with attempts, response, latency, signature and processing outcome.
- Configure retry/backoff, retryable errors, expiry, dead-letter queue, replay authority and alert thresholds.
- Reconcile consumer acknowledgment or downstream state with Seat Inventory and record compensating action.
- All writes require authenticated, authorized, idempotent and concurrency-safe requests; events and audit records carry tenant, version and correlation context and are retained under approved policy.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 09 - Immutable Seat Audit Logs

Purpose: Provide searchable tamper-evident evidence for seat data and administrative actions.

- Record actor/client, tenant, venue, event, entity, action, before/after, reason, source IP/device and timestamp.
- Store API request/event correlation, rule/version, approval, outcome and related transaction references.
- Support permission-controlled search and export, retention/legal hold and integrity verification without log alteration.
- All writes require authenticated, authorized, idempotent and concurrency-safe requests; events and audit records carry tenant, version and correlation context and are retained under approved policy.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Screen 10 - Integration Approval & Compliance

Purpose: Govern production activation and ongoing compliance of each integration.

- Route security review, data classification, privacy, architecture, performance and business-owner approval.
- Attach sandbox results, contract tests, load/security tests, rollback plan, support ownership and incident contacts.
- Issue time-bound approval, require periodic review and export a controlled compliance evidence package.
- All writes require authenticated, authorized, idempotent and concurrency-safe requests; events and audit records carry tenant, version and correlation context and are retained under approved policy.

Acceptance condition: Authorized users can complete the described task end to end; saved changes are validated, permission-controlled, integrated with the named shared services and traceable in the immutable audit history.

Implementation Handover Checklist

Check	Area	Required Evidence
☐	Functional	All 130 screens are mapped to user stories, acceptance criteria and traceable matrix requirements.
☐	Architecture	One Seat Inventory Engine and shared platform services are reused; no duplicate seat-state logic is introduced by channel or module.
☐	Data	Venue geometry, layout versions, identifiers, seat states, migration, retention, event ordering and reconciliation are documented.
☐	Security	RBAC/PBAC, SSO/MFA, tenant isolation, masking, SoD, temporary access, secrets, override and audit requirements are tested.
☐	Concurrency	Lock, hold, reservation, sale, release, exchange, idempotency, version and oversell protections are proven at peak load.
☐	Integrations	APIs, events, webhooks, schemas, retries, limits, correlation, monitoring, backward compatibility and reconciliation are validated.
☐	AI	Models/use cases, data access, confidence, explainability, human review, bias/accessibility checks, monitoring and governance are complete.
☐	UX & accessibility	Approved TICVAI design system, responsive behavior, keyboard support and WCAG 2.2 AA-aligned checks are complete.
☐	Quality	Unit, integration, end-to-end, concurrency, performance, security, accessibility, localization and UAT evidence is available.
☐	Operations	SLOs, dashboards, alerts, logs, runbooks, support ownership, backup/recovery and incident procedures are ready.
☐	Handover	Source code, configuration, documentation, test assets, deployment instructions and knowledge transfer are complete.

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